INSTRUCTIONS

AF3060 INVERTER DRIVES

DRIVER - 331X351BAG01, G02

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62513A

REV. 1 6/21/77

DRIVER ASSEMBLY	ELEM	36D865532BA
	CONN	36C764841BA

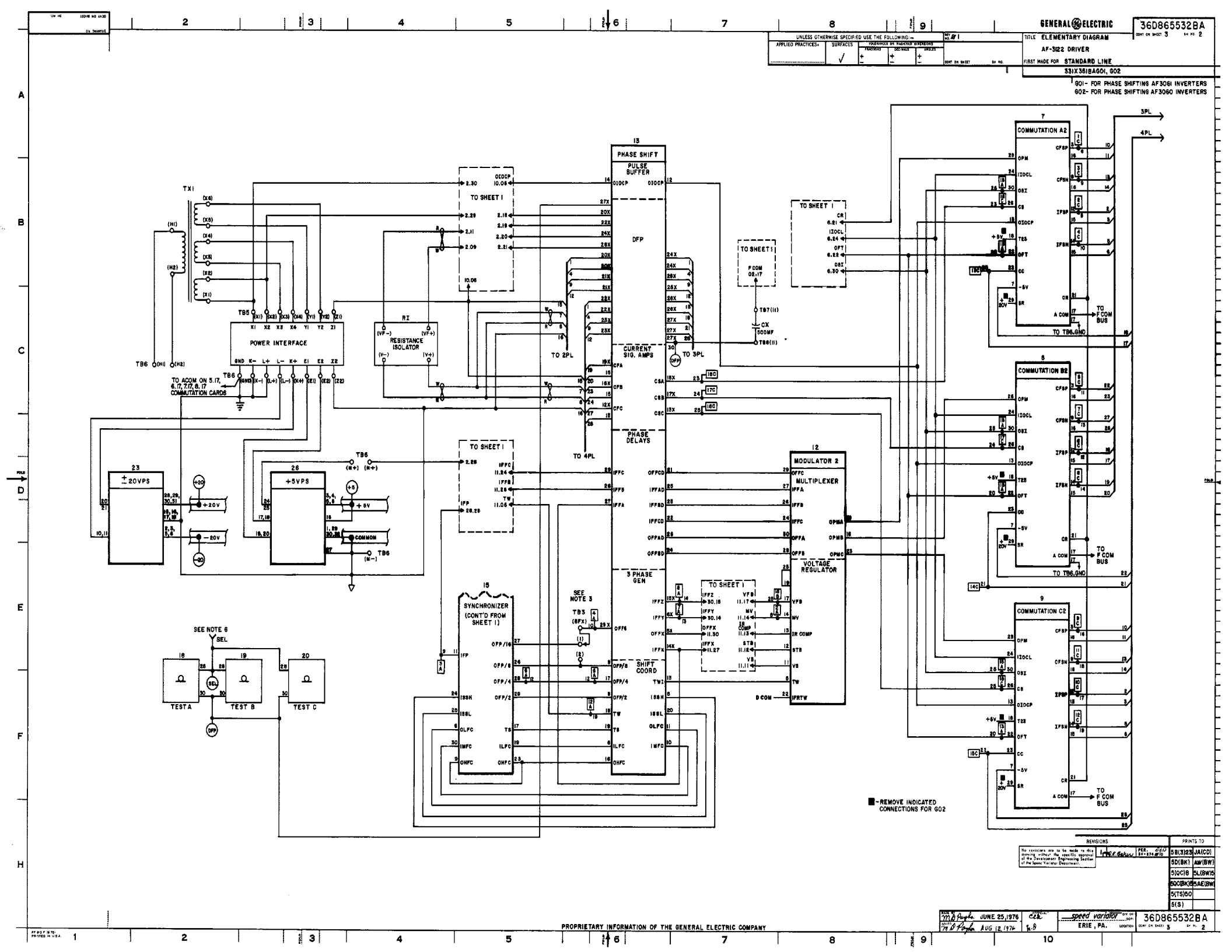
PRINTED CIRCUIT CARD DIAGRAMS

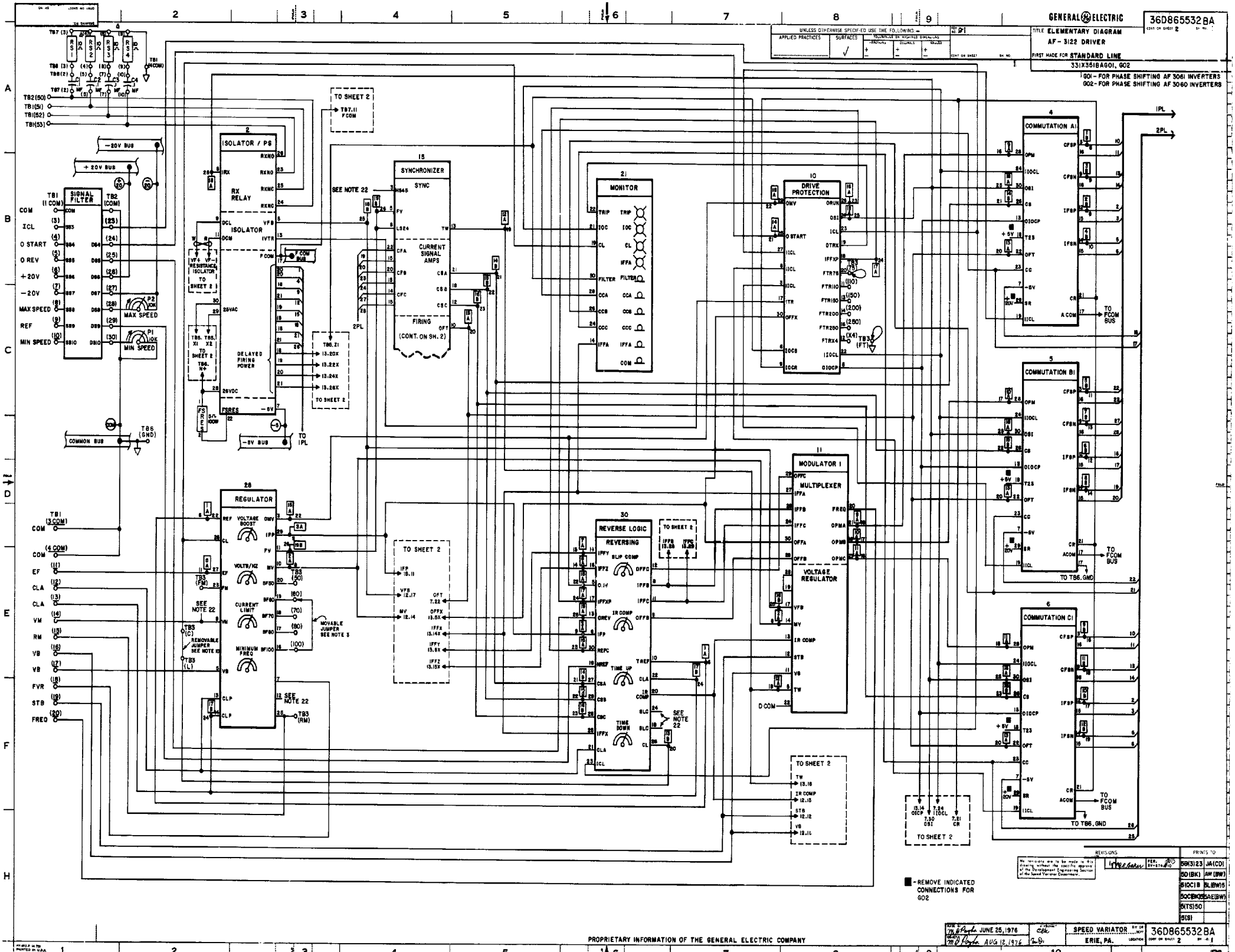
DRIVE PROTECTION	36C762849AC
COMMUTATION	36C764126AC
REGULATOR	36C764136AB
MODULATOR	36C764107AA
SYNCHRONIZER	36C762853AC
REVERSE LOGIC	36C762855AB
SIGNAL FILTER	36C762868AA
MONITOR	36C762854AA
ISOLATOR/POWER SUPPLY	36C762869CA
5V POWER SUPPLY	36C762870AB
20V POWER SUPPLY	36C762896AB
TEST CARD	36C762858AA
POWER INTERFACE	36C764129AA
RESISTANCE ISOLATOR	36C764140AA
PHASE SHIFT	36C762893CA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company

GENERAL ELECTRIC

demark of the General Electric Co.


GENERAL ELECTRIC
36D865532BA
TITLE ELEMENTARY DIAGRAM
AF-322 DRIVER
FIRST MADE FOR STANDARD LINE
331X381BAG01, G02
G01- FOR PHASE SHIFTING AF3061 INVERTERS
G02- FOR PHASE SHIFTING AF3060 INVERTERS
PHASE SHIFT
PULSE BUFFER
DFP
CURRENT SIG. AMPS
PHASE DELAYS
3 PHASE GEN
SHIFT COORD
VOLTAGE REGULATOR
MODULATOR 2
MULTIPLEXER
SYNCHRONIZER (CONT'D FROM SHEET 1)
POWER INTERFACE
RESISTANCE ISOLATOR
±20VPS
+5VPS
COMMUTATION A2
COMMUTATION B2
COMMUTATION C2
TO ACOM ON 5.17, 6.17, 7.17, 8.17 COMMUTATION CARDS
TO 2PL
TO 3PL
TO 4PL
TO SHEET 1
TO F COM BUS
TO TB6,GND
SEE NOTE 3
SEE NOTE 6
TEST A
TEST B
TEST C
■-REMOVE INDICATED CONNECTIONS FOR G02
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
JUNE 25,1976
AUG 12,1976
speed variator
ERIE, PA.
36D865532BA


GENERAL ELECTRIC
36D865532BA
TITLE ELEMENTARY DIAGRAM
AF-3122 DRIVER
FIRST MADE FOR STANDARD LINE
33(X361BAG01, G02
G01 - FOR PHASE SHIFTING AF 3061 INVERTERS
G02 - FOR PHASE SHIFTING AF 3060 INVERTERS
SIGNAL FILTER
ISOLATOR / PS
RX RELAY
ISOLATOR
SYNCHRONIZER
MONITOR
DRIVE PROTECTION
COMMUTATION AI
COMMUTATION BI
COMMUTATION CI
REGULATOR
REVERSE LOGIC
REVERSING
MODULATOR I
MULTIPLEXER
VOLTAGE REGULATOR
CURRENT SIGNAL AMPS
FIRING
DELAYED FIRING POWER
MAX SPEED
MIN SPEED
COMMON BUS
-20V BUS
+20V BUS
-8V BUS
TO IPL
TO FCOM BUS
TO SHEET 2
SEE NOTE 22
REMOVABLE JUMPER SEE NOTE 10
MOVABLE JUMPER SEE NOTE 5
■ - REMOVE INDICATED CONNECTIONS FOR G02
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE, PA.
36D865532BA
JUNE 25, 1976
AUG 12, 1976

WIRE TABLE

BUS BAR CONNECTIONS

TAB	CARD CONNECTED	CIRCUIT
02	02, 04, 05, 06, 07, 08, 09 10,11,12,13, 15	-20V
02	18, 19, 20, 21, 23, 26, 28,30	-20V
04	02, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13,15	+5V
04	18, 19, 20, 21, 23, 26, 28, 30	+5V
07	02, 04, 05, 06, 07, 08, 09, 10.	-5V
31	02, 04, 05, 06, 07,08,09,10,11,12,13,15	+20V
31	18,19,20,21,23,26,28,30	+20V
32	02,04,05,06,07,08,09,10,11,12,13,15	COM
32	10 11 12 13 15 BEND BUS BAR	COM
01	25 28 30	COM
17	02,04,05,06,07,08,09	F COM

BACK PLANE WIRING

#24 WHITE

19.25-02.05-11.17-12.17
18.25-02.06-04.30-05.30-06.30-07.30-08.30-09.30-10.24
02.13-10.17
02.18-13.20X
02.19-13.22X
02.20-13.24X
02.21-13.26X
04.03-19.06
04.09-19.09
04.12-19.08
04.13-05.13-06.13-10.05-13.14
04.14-19.10
04.18-04.04 ■
04.19-10.27
04.21-05-21-06.21-07.21-08.21-09.21-10.09
18.20-04.22-05.22-06.22-07.22-08.22-09.22-15.10
04.23-21.28
04.24-05.24-06.24-07.24-08.24-09.24-10.22-21.21
04.26-19.21-30.27-15.21
18.16-04.28-11.21
04.29-04.31 ■

#24 WHITE

05.03-19.11
05.09-19.13
05.12-19.12
05.14-19.14
05.18-05.04 ■
05.19-10.13
05.23-21.26
19.22-05.26-15.18-30.29
18.17-05.28-11.18
05.29-05.31 ■
06.03-19.16
06.09-19.18
06.12-19.17
06.14-19.19
06.18-06.04 ■
06.19-10.08
06.23-21.24
19.23-06.26-15.12-30.28
18.18-06.28-11.23
06.29-06.31 ■

07.03-20.06
07.09-20.09
07.12-20.08
07.13-08.13-09.13-13.12
07.14-20.10
07.18-07.04 ■
07.23-20.20
20.23-07.26-13.18X
07.28-12.21
07.29-07.31 ■

24 WHITE

08.03-20.11
08.09-20.13
08.12-20.12
08.14-20.14
08.18-08.04 ■
08.23-20.21
20.24-08.26-13.17X
08.28-12.18
08.29-08.31 ■

09.03-20.16
09.09-20.18
09.12-20.17
09.14-20.19
09.18-09.04 ■
09.23-20.22
20.25-09.26-13.13X
09.28-12.23
09.29-09.31 ■

#24 WHITE (CONT'D)

10.06-21.30
10.18-18.24-30.17
10.19-21.22
10.23-30.23
10.25-18.21
10.26-18.23-30.30
10.29-18.22-28.03-30.05
10.30-11.30-13.05X
21.14-11.27-13.14X-13.27-30.25
18.19-11.05-13.18-15.13
11.11-12.11
11.12-12.12
11.13-12.13-30.20
18.08-11.14-12.14-28.10
11.19-11.25
11.22-11.01
11.24-13.29-30.11
11.26-13.28-30.08
11.28-30.09
11.29-30.12

12.05-13.13
12.19-12.25
12.22-12.01
12.24-13.22
12.26-13.23
12.27-13.25
12.28-13.24
12.29-13.21
12.30-13.26

13.05-15.24
13.06-15.19
13.06X-30.14-18.13
13.08-15.29
13.09-15.26
13.10-15.30
13.11-15.06
13.15X-30.16-18.14
13.16-15.09-15.23
13.17-15.28-18.12
13.19-15.17
13.20-15.25
13.29X-18.10
15.05-15.08-28.11-19.26
18.09-28.29-30.06-15.11

18.06-28.22-30.10
18.11-28.27
18.26-30.13
18.28-19.28-20.28
18.30-19.30-20.30-13.27X
19.20-30.26
19.24-28.14-30.22
21.19-28.26
28.13-30.21

+5V

26.04-26.16
02.04-18.04
04.04-19.04
05.04-20.04
06.04-21.04

COMMON

26.01-26.27
23.15-11.32
23.16-12.32
23.17-13.32
23.18-28.01

F COM

02.17-18.01-19.01-20.01-21.01

-5V

02.07-18.07-19.07-20.07-21.07

-20V

02.02-18.02
04.02-19.02
05.02-20.02

+20V

02.31-18.31
04.31-19.31
05.31-20.31

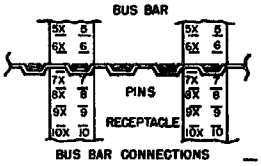
TOP VIEW RACK OPEN

2	4	5	6	7	8	9	10	11	12	13	15
ISOL/PS	COMM A1	COMM B1	COMM C1	COMM A2	COMM B2	COMM C2	PROT	MOD 1	MOD 2	SEE SK "A" AND NOTE 6 PHASE SHIFT	SYNC

18	19	20	21	23	26	28	30
TEST A	TEST B	TEST C	MONITOR	± 20V	+5V	REG	REV LOGIC

SKETCH "B"

BUS BAR



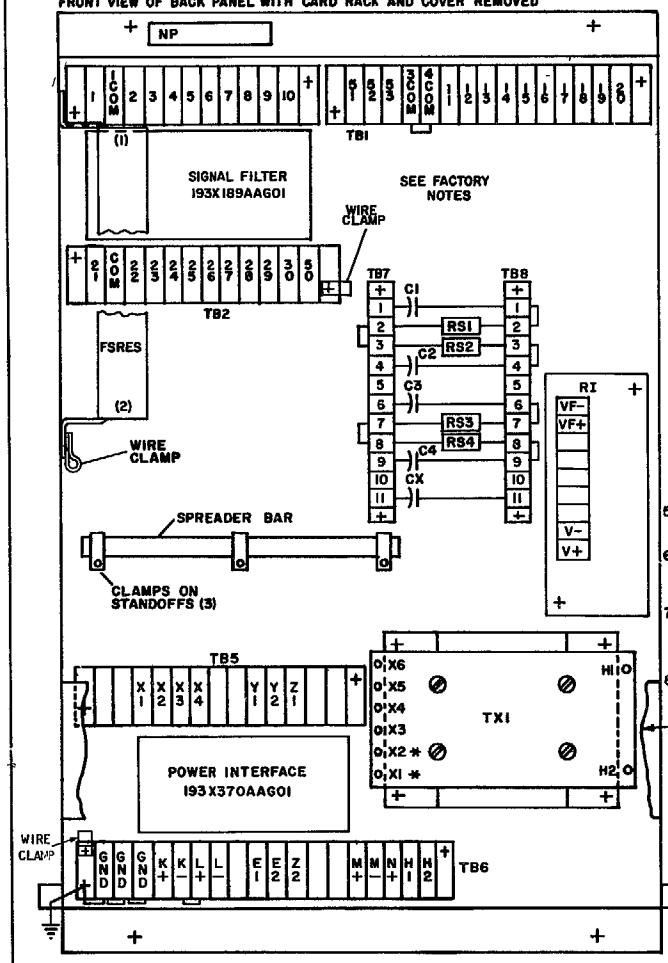
BUS BAR CONNECTIONS

FACTORY NOTES:

1. ASSEMBLE BUS BAR BEFORE WIRING. ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE.
2. CONNECT WIRES TO TB1, TB2, & TB3 AT LOWER TERMINALS; ALL WIRES TO TB5 & TB6 AT UPPER TERMINALS.
3. ✱-CRIMP ALL #24 WIRES ON ONE SPADE TERMINAL.
4. TB3 HAS BOTH SCREW AND TERMIPOINT CONNECTORS.

TB3 + 50 60 70 80 100 BFX 1 2 + 75 110 150 200 250 FT X4 RM FM C L +

FRONT VIEW OF BACK PANEL WITH CARD RACK AND COVER REMOVED

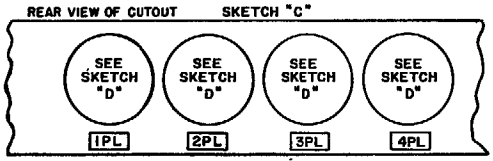


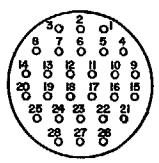
SKETCH "A"
FULL SIZE VIEW OF CARD RECEPTACLE

1X– – 1
2X– – 2
3X– – 3
4X– – 4
5X– – 5
6X– – 6
7X– – 7
8X– – 8
9X– – 9
10X– – 10
11X– – 11
12X– – 12
13X– – 13
14X– – 14
15X– – 15
16X– – 16
17X– – 17
18X– – 18
19X– – 19
20X– – 20
21X– – 21
22X– – 22
23X– – 23
24X– – 24
25X– – 25
26X– – 26 SEE NOTE 3 ON SH. 2
27X– – 27
28X– – 28
29X– – 29
30X– – 30
31X– – 31
32X– – 32

5. LOCATION 13 IS A 64 PIN RECEPTACLE ALL OTHERS ARE 32 PIN.
6. DO NOT BUNDLE WIRES. DISTRIBUTE WIRES ALONG LENGTH OF SPREADER BAR.
7. CUT ALL WIRES APPROXIMATELY ONE INCH LONGER THAN THE SHORTEST POSSIBLE LENGTH.
8. ■ REMOVE FOR G02 ONLY.

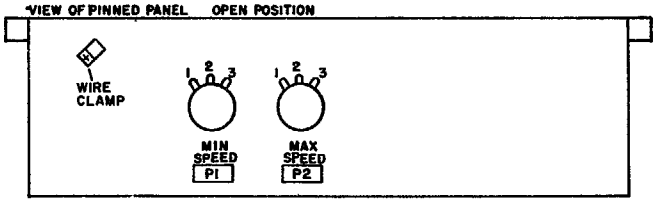
REAR VIEW OF CUTOUT SKETCH "C"





SKETCH "D"

VIEW OF PINNED PANEL OPEN POSITION



UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
FRACTIONS DECIMALS MACHINED ANGLES
DIMENSIONS ON
718.5503

TITLE CONNECTION DIAGRAM AF3122 DRIVER
FIRST MADE FOR STANDARD LINE 331X351BAG01, G02
G01 – FOR PHASE SHIFTING AF3061 INVERTERS
G02 – FOR PHASE SHIFTING AF3050 INVERTERS

GENERAL ELECTRIC
36C764841BA SH NO 1
CONT ON SHEET 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN Mike Boyle JUNE 28,1976
ENGR Fritz Stark AUG 12,1976
APPROVALS
SPEED VARIATOR DEPT
ERIE, PA
LOCATION
36C764841BA SH NO 2
LAST ON SHEET 2

PRINTS TO	JA(CD)	SB(13)	AW(BW)	SB(1323)	5L(BW)5	5D(BK)	SAE(BW)	BASC(B)	5D1980	5D2800

REVISIONS

REV NO	DESCRIPTION	DATE	APPROVED
1	Any revisions are to be made to this drawing only by authority of the Drive Control Engineering Section of the Speed Variator Department.	8-17-76	

DRAWING BY
CONT ON SHEET
SH NO

WIRE TABLE

INTERFACE WIRING
#24 WHITE

P1.01-TB2.30
P1.02-P1.03-TB6.GND
P2.01-P2.02-TB2.28
P2.03-TB2.27
TB2.5D-02.24
TB1.51-02.25
TB1.52-02.23
TB1.53-02.26

TB1.4 COM-TB1.3 COM
TB1.11-28.27
TB1.12-30.21
TB1.13-30.22
TB1.14-28.09
TB1.15-28.25
TB1.16-11.11
TB1.17-28.05
TB1.18-28.07
TB1.19-11.12
TB1.20-11.20
TB2.23-10.23
TB2.24-10.25
TB2.25-30.13
TB2.26-23.29
TB2.27-23.02
TB2.29-30.19

TB3.75-10.20
TB3.110-10.11
TB3.150-10.13
TB3.200-10.14
TB3.2-15.26
TB3.1-15.27
TB3.BFX-13.29X

TB3.50-28.20
TB3.60-28.19
TB3.70-28.18
TB3.80-28.17
TB3.100-28.16

TB3.L-30.26
TB3.C-28.26
TB3.250-10.16
TB3.X4-10.12
TB3.FT-10.15
TB3.RM-28.25
TB3.FM-28.23

TB5.Z1-10.06

TB6.M(MINUS)-26.29*
TB6.M(MINUS)-26.30*
TB6.M(MINUS)-26.32*
TB6.L(MINUS)-23.10*
TB6.L(MINUS)-23.10*
TB6.L(MINUS)-23.11*
TB6.K(PLUS)-23.20*
TB6.K(PLUS)-23.20*
TB6.K(PLUS)-23.21*
TB6.M(PLUS)-26.24*
TB6.M(PLUS)-26.24*
TB6.M(PLUS)-26.25*
TB6.E1-26.19*
TB6.E1-26.19*
TB6.E1-26.20*
TB6.E2-26.17*
TB6.E2-26.17*
TB6.E2-26.18*
TB6.N(PLUS)-02.28*
TB6.N(PLUS)-02.28*
TB6.GND-05.17*
TB6.GND-06.17*
TB6.GND-07.17*
TB6.GND-08.17*

TX1.X1-02.30*
TX1.X1-02.30*
TX1.X1-02.30*

TX1.X2-02.29*
TX1.X2-02.29*
TX1.X2-02.29*

TB2.50-TB7.1
TB1.51-TB7.4
TB1.52-TB7.6
TB1.53-TB7.9
TB1.4 COM-TB7.2
TB7.3-TB7-7

18.01-TB7.11
18.30-TB8-11

#14 BLACK

TX1.X1-TB5.X1
TX1.X2-TB5.X2
TX1.X3-TB5.X3
TX1.X4-TB5.X4
TX1.X5-TB5.Y1
TX1.X6-TB5.Y2
TX1.H1-TB6.H1
TX1.H2-TB6.H2
TB6.GND-TB2.COM-TB1.3COM
TB6.GND-TB6.K MINUS
TB6.L(PLUS)-TB6.Z2

#24 RED & WHITE TWISTED

02.09-R1 .VF+ WHITE
02.11-R1 .VF- RED
2PL.07-R1 .V+ WHITE
2PL.08-R1 .V- RED
4PL.07-R1 .V+ WHITE
4PL.08-R1 .V- RED

PLUG WIRING
#24 WHITE

1PL.01-02.20
1PL.02-05.12
1PL.03-04.15
1PL.04-02.20
1PL.05-04.14
1PL.06-04.15
1PL.09-02.18
1PL.10-04.03
1PL.11-04.16
1PL.12-02.21
1PL.13-04.09
1PL.14-04.16
1PL.15-02.19
1PL.16-05.12
1PL.17-05.15
1PL.18-02.19
1PL.19-05.14
1PL.20-05.15
1PL.21-02.18
1PL.22-05.03
1PL.23-05.16
1PL.26-02.21
1PL.27-05.09
1PL.28-05.16

2PL.01-13.20X
2PL.02-06.12
2PL.03-06.15
2PL.04-13.20X
2PL.05-06.14
2PL.06-06.15

2PL.09-13.21X
2PL.10-06.03
2PL.11-06.16
2PL.12-13.21X
2PL.13-06.09
2PL.14-06.16
2PL.15-21.30
2PL.16-10.32
2PL.17-04.23
2PL.18-04.07
2PL.19-15.22
2PL.20-15.15
2PL.21-05.23
2PL.22-05.07
2PL.23-15.20
2PL.24-15.15
2PL.25-06.23
2PL.26-06.07
2PL.27-15.14
2PL.28-15.15

3PL.01-13.24X
3PL.02-07.12
3PL.03-07.15
3PL.04-13.24X
3PL.05-07.14
3PL.06-07.15

FS RES.1-02.28
FS RES.2-02.22

3PL.09-13.25X
3PL.10-07.03
3PL.11-07.16
3PL.12-13.25X
3PL.13-07.09
3PL.14-07.16
3PL.15-13.26X
3PL.16-08.12
3PL.17-08.15
3PL.18-13.26X
3PL.19-08.14
3PL.20-08.15
3PL.21-13.27X
3PL.22-08.03
3PL.23-08.16
3PL.26-13.27X
3PL.27-08.09
3PL.28-08.16

4PL.01-13.22X
4PL.02-09.12
4PL.03-09.15
4PL.04-13.22X
4PL.05-09.14
4PL.06-09.15

4PL.09-13.23X
4PL.10-09.03
4PL.11-09.16
4PL.12-13.23X
4PL.13-09.09
4PL.14-09.16
4PL.15-21.30
4PL.16-10.01
4PL.17-07.23
4PL.18-07.07
4PL.19-13.19X
4PL.20-13.15
4PL.21-08.23
4PL.22-08.07
4PL.23-13.16X
4PL.24-13.15
4PL.25-09.23
4PL.26-09.07
4PL.27-13.12X
4PL.28-13.15

FRONT VIEW OF CARD RACK CLOSED POSITION

	18	19	20	21	25	26	28	30
LOCATION NUMBERS	193X188AAG01 (SEE NOTE 5)	193X188AA 601	193X188AA 601	193X184AA 601	193X257AAG01	193X191AB G01	193X 271AA 601	188X185AAG01
	TEST A	TEST B	TEST C	MON-ITOR	± 20VPS	+5VPS	REG	REV LOGIC

	2	4	5	6	7	8	9	10	11	12	13	15
LOCATION NUMBERS	188X190AFG02	193X 348AA 601	193X 348AA 601	193X 348AA 601	193X 349AA 601	188X 349AA 601	193X 349AA 601	193X 178AC 601	193X 197AA 601	193X 197AA 601	193X 194CA 601	193X188AC601
	ISOL/PS	COMM A1	COMM B1	COMM C1	COMM A2	COMM B2	COMM C2	PRO-TECT-ION	MOD-ULATOR 1	MOD-ULATOR 2	PHASE SHIFT	SYN-CHRONIZER

NOTES:

1. LOCATION NUMBERS AT THE TOP OF EACH CARD INDICATE THE SPACE NUMBER THAT THE CARD'S RECEPTACLE OCCUPIES. THE CARD OCCUPIES THAT SAME SPACE AND AN INTEGRAL NUMBER OF SPACES TO THE LEFT. THE SPACE UNIT REQUIREMENT FOR A CARD CAN BE DETERMINED BY SUBTRACTING THE SPACE NUMBER OF THE CARD TO THE LEFT.

2. THE WIRE TABLE LISTS ALL WIRE CONNECTIONS IN THE DRIVER. A CONNECTION IS INDICATED BY A CARD LOCATION NUMBER, AND THE RECEPTACLE PIN NUMBER SEPARATED BY A PERIOD (E.G. 02.30 INDICATES LOCATION 2, PIN 30). OTHER CONNECTIONS SUCH AS TO A TERMINAL BOARD ARE INDICATED BY THE DEVICE NOMENCLATURE AND A POINT NUMBER (E.G. TB.01). A WIRE BETWEEN TWO POINTS IS INDICATED BY A DASH BETWEEN THE TWO CONNECTION POINT NOMENCLATURES (E.G. TB.21 - 23.06). A CONNECTION MADE BY BUS BAR AS SHOWN IN SKETCH "B" IS INDICATED IN THE TABLE AT THE BEGINNING OF THE WIRE TABLE.

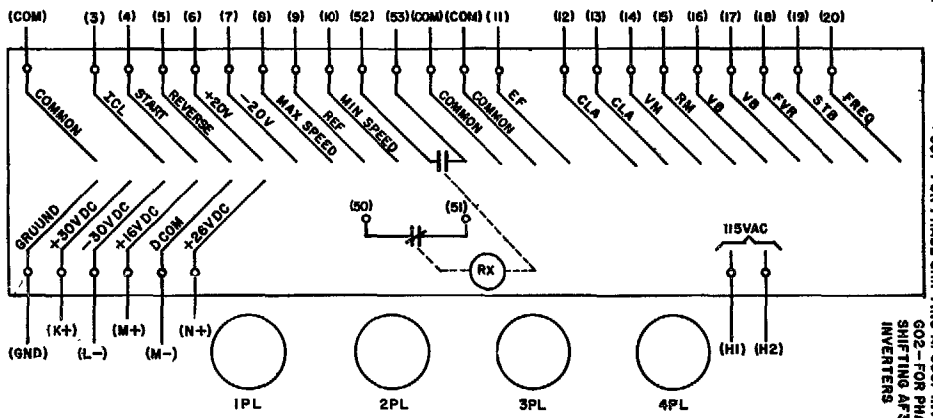
3. THE RECEPTACLE IS NUMBERED CORRESPONDING TO ITS INDIVIDUAL PINS. CARDS HAVE TAB NUMBERS WHICH CORRESPOND TO PIN NUMBERS ON THE RECEPTACLE. FOR A CARD WITH TABS ON BOTH SIDES THE CORRESPONDING RECEPTACLE HAS 64 INDIVIDUAL PINS. A CARD WITH TABS ON ONE SIDE HAS A CORRESPONDING RECEPTACLE WITH 32 PINS.

4. ON PRINTED CIRCUIT CARDS USED IN THIS REGULATOR THE LETTERS "AA" AFTER THE BASIC CARD CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBERS AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS AB, AC, AD, ETC., ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN "AA" CANNOT REPLACE AN "AB"

5. LOCATION NUMBERS 19 & 20 HAVE NO PERMANENT CARD, IT IS A TEST RECEPTACLE AND MAY BE USED WITH THE TEST CARD IN LOCATION NUMBER 18.

MOVEABLE JUMPER CONNECTIONS	MOVEABLE JUMPER CONNECTIONS
THE FIRST WIRE CONNECTION NUMBER IS THE PERMANENT ONE. THE SECOND WIRE CONNECTION NUMBER IS THE MOVEABLE ONE. #24 WHITE 19.28-19.28	THE FIRST WIRE CONNECTION BELOW IS THE PERMANENT TERMIPOINT CONNECTION. THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE USING A SPADE TERMINAL. #24 RED TB3.100-TB3.60 TB3.BFX-TB3.01 TB3.75-TB3.75 TB3.FT-TB3.FT } WIRES SHOULD BE 3.5 INCHES IN LENGTH

DRIVER SUBASSEMBLY EXTERNAL CONNECTIONS



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

SPEED VARIATOR — ERIE, PA

36C764841BA

REVISIONS

REV NO	DESCRIPTION	DATE	APPROVED
	Updated drawing	JUNE 28 1976	

No revisions are to be made to this drawing without approval of the Drive Development Engineering Section of the Speed Variator Department.

PRINTS TO

JA(CD) 51(B)	JW (DRW)	88(3125	
6LBW15	80G(BK)		
6AE(BK)	80G(BK)		
80G(BK)			
8XT(BK)			

GENERAL ELECTRIC

TITLE CONNECTION DIAGRAM
AF-3122 DRIVER
FIRST MADE FOR STANDARD LINE
531X351BAG01, G02
I G01 — FOR PHASE SHIFTING AF3061 INVERTERS
G02 — FOR PHASE SHIFTING AF3060 INVERTERS

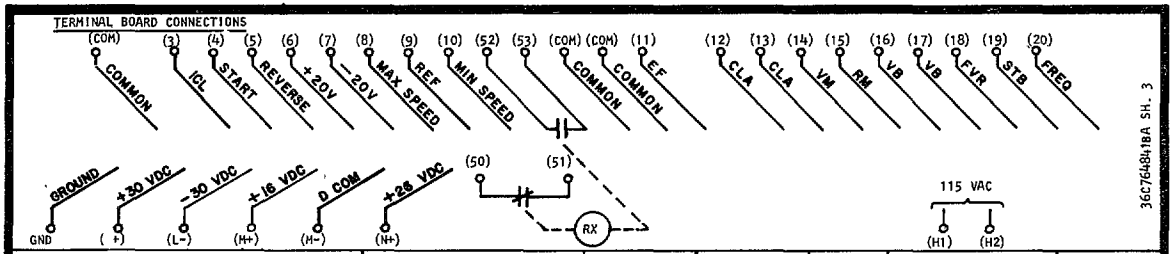
UNLESS OTHERWISE SPECIFIED, USE THE FOLLOWING —
APPROX. FRACTIONS ± 1/32
TOLERANCES ON MACHINED DIMENSIONS: DECIMALS ± .005, ANGLES ± 1°
SURFACES √

719A505

36C764841BA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE LABEL ARTWORK & NOTES
AF 3122 DRIVER
FIRST MADE FOR STANDARD LINE
331X351BAG01,G02



TEST CARD

THE TEST CARD MAY BE PLACED IN EITHER RECEPTACLE A,B,OR C A MOVEABLE JUMPER CONNECTED TO PIN 28 OF THE TEST RECEPTACLES MAY BE USED TO MONITOR ANY POINT IN THE BACK PLANE THROUGH THE "SEL" TEST CONNECTOR ON THE TEST CARDS.

TEST POINTS

	A		B		C	
1.	REF	MASTER	CFSP	SLAVE	CFSP	A
2.	MV		IFSP		IFSP	
3.	1FP		CFSN		CFSN	
4.	OFF6		IFSN		IFSN	
5.	EF		CFSP		CFSP	B
6.	OFP/4		IFSP		IFSP	
7.	1FFY		CFSN		CFSN	
8.	1FFZ		IFSN		IFSN	
9.	OMPA		CFSP		CFSP	C
10.	OMPB		IFSP		IFSP	
11.	OMPC		CFSN		CFSN	
12.	TW		IFSN		IFSN	
13.	OFT		CL		CCA	
14.	OSTART		CSA		CCB	
15.	OMV		CSB		CCC	
16.	ORUN		CSC		CSA	
17.	1FEXP		CLA		CSB	
18.	OS1		VFB		CSC	
19.	OREV		FV			

+20V, DFP, SEL, −5V, +5V, −20V, 10, READ, COM

MONITOR CARD

DRIVER TRIP (IOC WILL ALSO LIGHT)
INSTANTANEOUS OVERCURRENT TRIP
CURRENT LIMIT ON
FREQUENCY SIGNAL
FILTER F.B.
CCA
CCB
CCC
1FFA
COMMON

REVERSE LOGIC

P475 SLIP COMP
P476 IR COMP
P477 TIME UP
P478 TIME DOWN

REGULATOR

P265 VOLTAGE BOOST
P267 V/HZ
P266 CURRENT LIMIT
P268 MIN FREQ.

SPACE NUMBER	NAME	CARD NUMBER
2	ISOLATOR/PS	193X190BAG02
4	COMMUTATION "A"	193X369ABG01
5	COMMUTATION "B"	193X369ABG01
6	COMMUTATION "C"	193X369ABG01
7	COMMUTATION "A"	193X369ABG01
8	COMMUTATION "B"	193X369ABG01
9	COMMUTATION "C"	193X369ABG01
10	DRIVE PROTECTION	193X179ACG01
11	MODULATOR 1	193X197AAG01
12	MODULATOR 2	193X197AAG01
13	PHASE SHIFT	193X194CAG01
15	SYNCHRONIZER	193X183ACG01
18	TEST A	193X188AAG01
19	TEST B	193X188AAG01
20	TEST C	193X188AAG01
21	MONITOR	193X184AAG01
23	±20V PS	193X257AAG01
26	+5V PS	193X191ABG01
28	REGULATOR	193X371ABG01
30	REVERSE LOGIC	193X185AAG01

POWER SUPPLY CARDS

BEFORE REPLACING FUSES(S) ON POWER SUPPLY CARDS, CHECK FOR POSSIBLE SHORT(S) ACROSS THE OUTPUT(S). REPLACE WITH THE SAME TYPE AND RATING (SPARE FUSE MOUNTED ON THE CARD.

SPACE FOR CAUTION LABEL
N P. 104X905BA148

FREQUENCY TRIP SELECTION

TRIP FREQUENCY IS SELECTED BY A JUMPER ON TB3 AS FOLLOWS

TB3 (75) JUMPER	75	110	150	200	250
FREQUENCY TRIP (HZ)	75	110	150	200	250
TB3 (FT) JUMPERED TO (X4)	300	440	600	800	1000

SPACE FOR WARNING LABEL
NP 104X905BA068

BASE FREQUENCY SECECTION

BASE FREQUENCY IS SELECTED BY TWO JUMPERS ON TB3 AS FOLLOWS

TB3 JUMPERS BFX \ HERTZ	(40)	(50)	(60)	(70)	(80)	(100)
(100) (1)	40	50	60	70	80	100
(2)	80	100	120	140	160	200

OUTPUT FREQ

SPACE FOR CAUTION LABEL
N.P. 104X905BA067

DRIVER DOOR LABEL

1 SIZE AS INDICATED.

2 PRINT ON FASSON WHITE CONVENTION BADGE (LAPEL) STOCK NO. 65P

3. CUT TO SIZE.

11"

13"

REV NO 0

REVISIONS

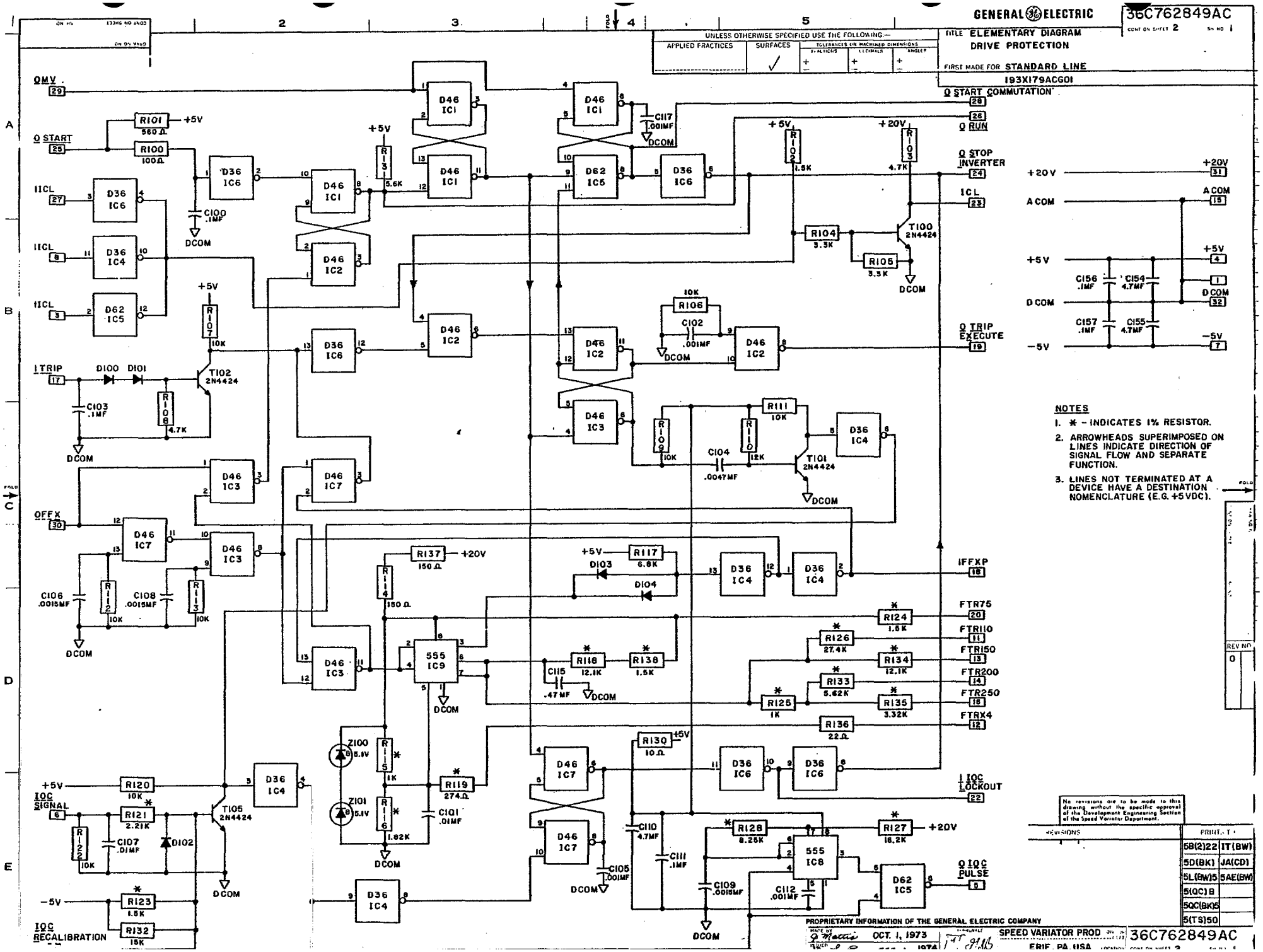
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5(S1)	JA(CD)
5B(3)23	AW (BW)
5D(BK)	5L(BW)S
5QC(B)	
5QC(BK)S	
5TS 50	

MADE M.D. Boyle JUNE 25, 1976
PROJECT M.D. Boyle AUG 12, 1976
ENGINEER E.E.E. 2.9.
SPEED VARIATOR ERIE, PA.
36C764841BA
CONT ON SHEET FL SH NO 3

FP-909-F (1-72) PRINTED IN U.S.A.


GENERAL ELECTRIC
36C762849AC
TITLE ELEMENTARY DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179ACG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
QMV 29
Q START 25
IICL 27
IICL 9
IICL 3
ITRIP 17
OFFX 30
IOC SIGNAL 6
IOC RECALIBRATION
Q START COMMUTATION 28
Q RUN 26
Q STOP INVERTER 24
ICL 23
Q TRIP EXECUTE 19
IFFXP 18
FTR75 20
FTRI10 11
FTRI50 13
FTR200 14
FTR250 15
FTRX4 12
I IOC LOCKOUT 22
Q IOC PULSE 5
+20V 31
ACOM 15
+5V 4
DCOM 32
-5V 7
R101 560Ω
R100 100Ω
R102 1.5K
R103 4.7K
R104 3.3K
R105 3.3K
R106 10K
R107 10K
R108 4.7K
R109 10K
R110 12K
R111 10K
R112 10K
R113 5.6K
R114 150Ω
R115 1K
R116 1.82K
R117 6.8K
R118 12.1K
R119 274Ω
R120 10K
R121 2.21K
R122 10K
R123 1.5K
R124 1.5K
R125 1K
R126 27.4K
R127 16.2K
R128 8.25K
R130 10Ω
R132 15K
R133 5.62K
R134 12.1K
R135 3.32K
R136 22Ω
R137 150Ω
R138 1.5K
C100 .1MF
C101 .01MF
C102 .001MF
C103 .1MF
C104 .0047MF
C105 .001MF
C106 .0015MF
C107 .01MF
C108 .0015MF
C109 .0015MF
C110 4.7MF
C111 .1MF
C112 .001MF
C115 .47MF
C117 .001MF
C154 4.7MF
C155 4.7MF
C156 .1MF
C157 .1MF
D100 DIOI
D102
D103
D104
Z100 5.1V
Z101 5.1V
T100 2N4424
T101 2N4424
T102 2N4424
T105 2N4424
555 IC9
555 IC8
D46 IC1
D46 IC2
D46 IC3
D46 IC7
D36 IC6
D36 IC4
D62 IC5
DCOM
NOTES
1. * - INDICATES 1% RESISTOR.
2. ARROWHEADS SUPERIMPOSED ON LINES INDICATE DIRECTION OF SIGNAL FLOW AND SEPARATE FUNCTION.
3. LINES NOT TERMINATED AT A DEVICE HAVE A DESTINATION NOMENCLATURE (E.G. +5VDC).
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
REVISIONS
PRINT
5B(2)22 1T(BW)
5D(BK) 1A(CD)
5L(BW)5 5AE(BW)
5(QC)8
5QC(BK)5
5(TS)50
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
MADE BY G. Martin OCT. 1, 1973
SPEED VARIATOR PROD
ERIE, PA. U.S.A.
36C762849AC

GENERAL ELECTRIC

36C762849AC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

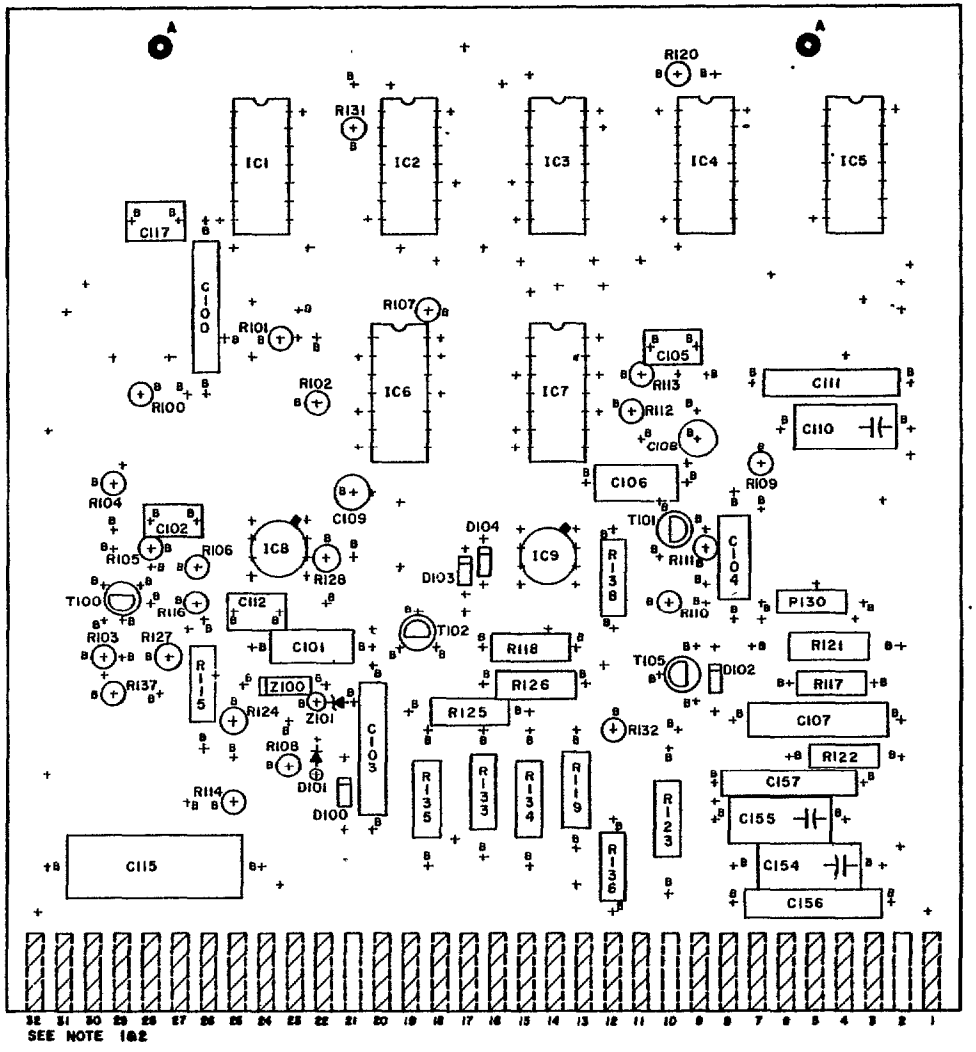
APPLIED PRACTICES	SURFACES	TOLERANCES OR MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
710A818	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179AC601

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

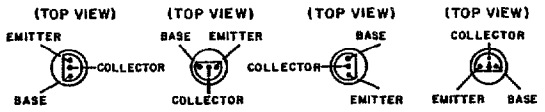
LOC.	DIA.	QUAN.
A	.157	2
B	.040	130

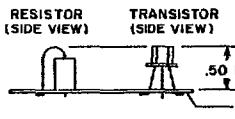


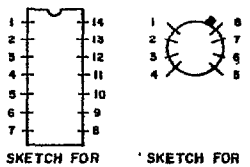
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.006}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

TRANSISTOR LEAD SKETCHES







2X	36A352983CA005	36A352982CA005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varistor Department

PRINTS TO

8QC(B)	5 (B)
58(2)22	6D(BK)
5L(BW)5	5D(B)
5QC(BK)5	5AE(BW)
5(TS)50	JA(CD)
1T(BW)	

REV NO. 0

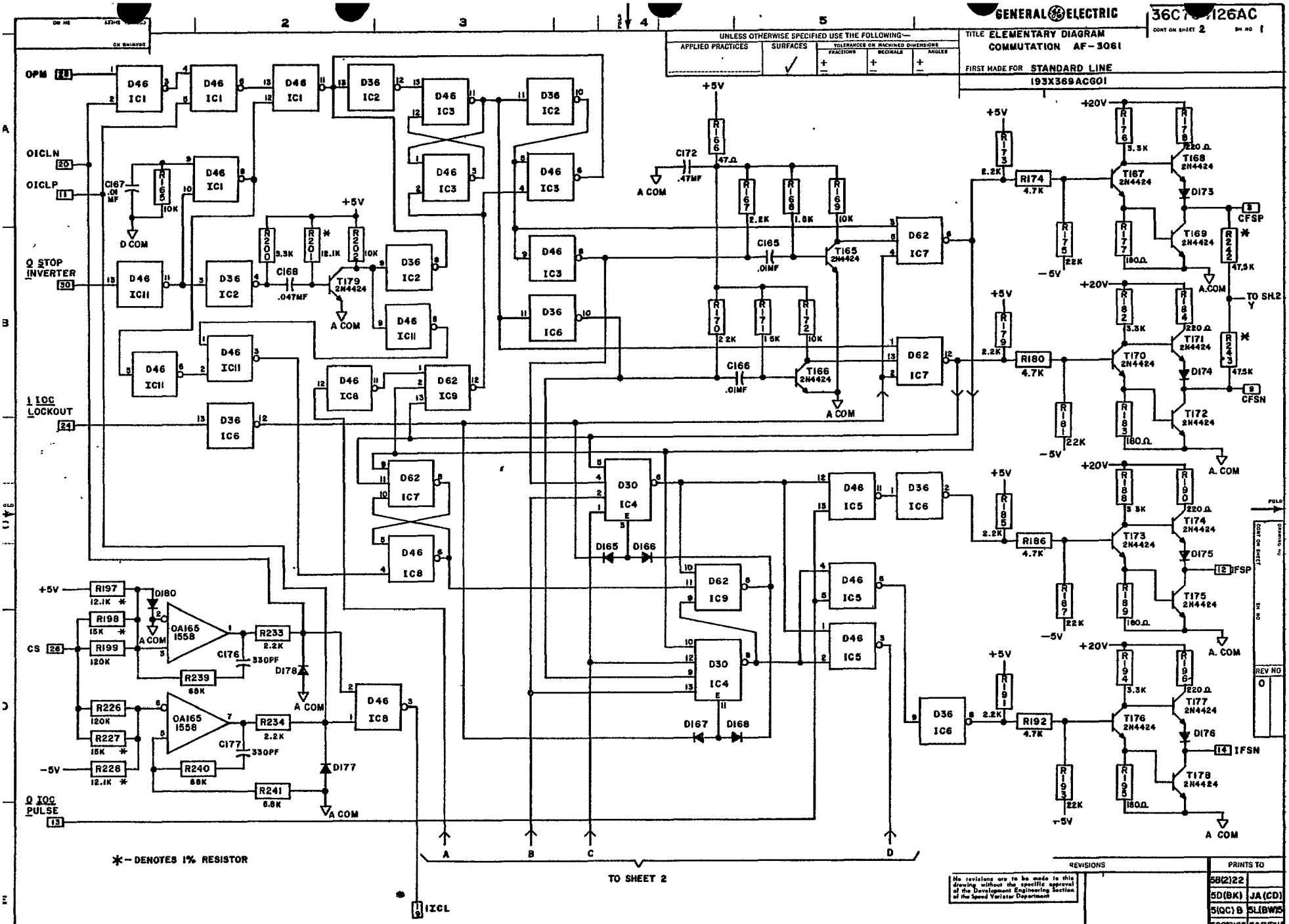
DRAWN BY G. Mattis OCT. 3, 1973 APPROVED

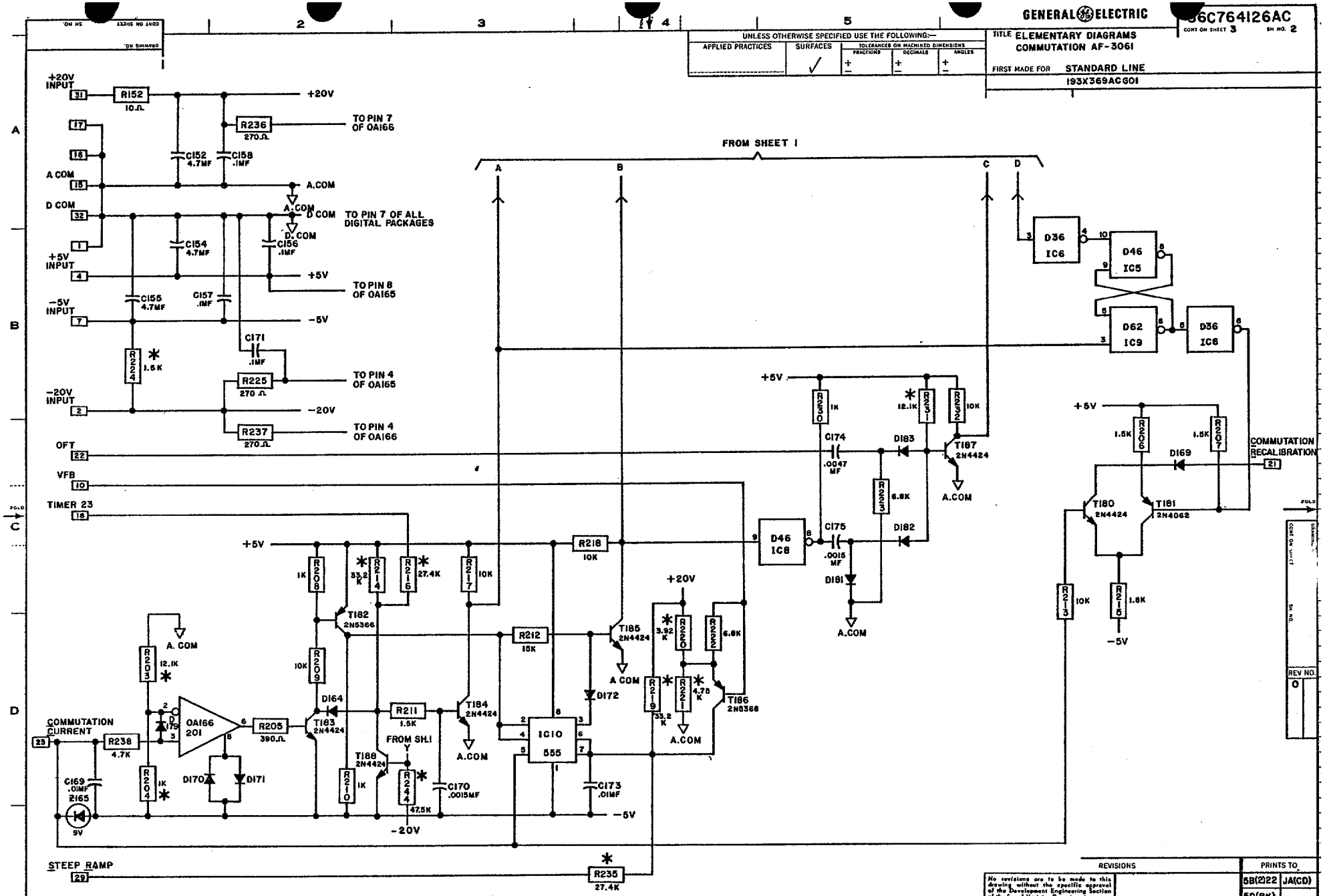
SPEED VARIATOR PROD DIV OR DEPT

ERIE, PA. USA LOCATION

36C762849AC CONT ON SHEET FL SH NO 2

GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764126AC
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01
OPM 23
OICLN 20
OICLP 11
O STOP INVERTER 50
1 IOC LOCKOUT 24
CS 26
O IOC PULSE 13
+5V
-5V
+20V
D COM
A COM
A. COM
D46 IC1
D46 IC2
D46 IC3
D36 IC2
D36 IC6
D46 IC11
D46 IC8
D62 IC7
D62 IC9
D30 IC4
D46 IC5
D36 IC6
CFSP 9
CFSN 9
IFSP 12
IFSN 14
TO SH.2
TO SHEET 2
IZCL
*—DENOTES 1% RESISTOR
REVISIONS
No revisions are to be made in this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department
PRINTS TO
58(2)22
5D(BK) JA(CD)
5(QC)B 5L(BW)5
5QC(BK)5 5AE(BW)
5(TS)50
5(S)
REV NO 0
MADE BY G.R. Baker MAY 25, 1977
ISSUED MAY 26,1977
APPROVALS
SPEED VARIATOR ERIE, PA
36C764126AC
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764126AC
CONT ON SHEET 3
SH NO. 2
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAMS
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01
+20V INPUT
R152 10Ω
+20V
R236 270Ω
TO PIN 7 OF OA166
C152 4.7MF
C158 .1MF
A COM
D COM
TO PIN 7 OF ALL DIGITAL PACKAGES
C154 4.7MF
C156 .1MF
+5V INPUT
+5V
TO PIN 8 OF OA165
C155 4.7MF
C157 .1MF
-5V INPUT
-5V
R224 1.5K
C171 .1MF
R225 270Ω
TO PIN 4 OF OA165
-20V INPUT
-20V
R237 270Ω
TO PIN 4 OF OA166
OFT
VFB
TIMER 23
FROM SHEET 1
COMMUTATION CURRENT
R238 4.7K
C169 .01MF
Z165 9V
OA166 201
R205 390Ω
R203 12.1K
R204 1K
DI70
DI71
DI79
R208 1K
R209 10K
R214 33.2K
R216 27.4K
R217 10K
T182 2N5366
DI64
T183 2N4424
R210 1K
T188 2N4424
R244 47.5K
C170 .0015MF
R211 1.5K
T184 2N4424
A.COM
R212 15K
R218 10K
T185 2N4424
DI72
IC10 555
C173 .01MF
R219 33.2K
R220 3.92K
R221 4.75K
R222 6.8K
T186 2N5366
D46 IC8
C174 .0047 MF
C175 .0015 MF
R220 1K
R221 12.1K
R222 10K
R223 6.8K
DI83
DI82
DI81
T187 2N4424
D36 IC6
D46 IC5
D62 IC9
D36 IC6
R206 1.5K
R207 1.5K
DI69
COMMUTATION RECALIBRATION
T180 2N4424
T181 2N4062
R213 10K
R215 1.8K
STEEP RAMP
R235 27.4K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(2)22 JA(CD)
5D(8K)
5(QC)8 5L(BW)5
5QC(BK)55AE(BW)
5(TS)50
5(S)
MADE BY R.R. Baker MAY 25, 1977
ISSUED R. Summer MAY 25, 1977
APPROVALS
speed variator DIV OR DEPT
ERIE, PA. LOCATION
36C764126AC
CONT ON SHEET 3 SH NO. 2
PREPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
FF-809-F (4-72)

GENERAL ELECTRIC

36C764126AC

CONT ON SHEET FL. SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A818	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
COMMUTATION AF-3061

FIRST MADE FOR STANDARD LINE

193X369ACG01

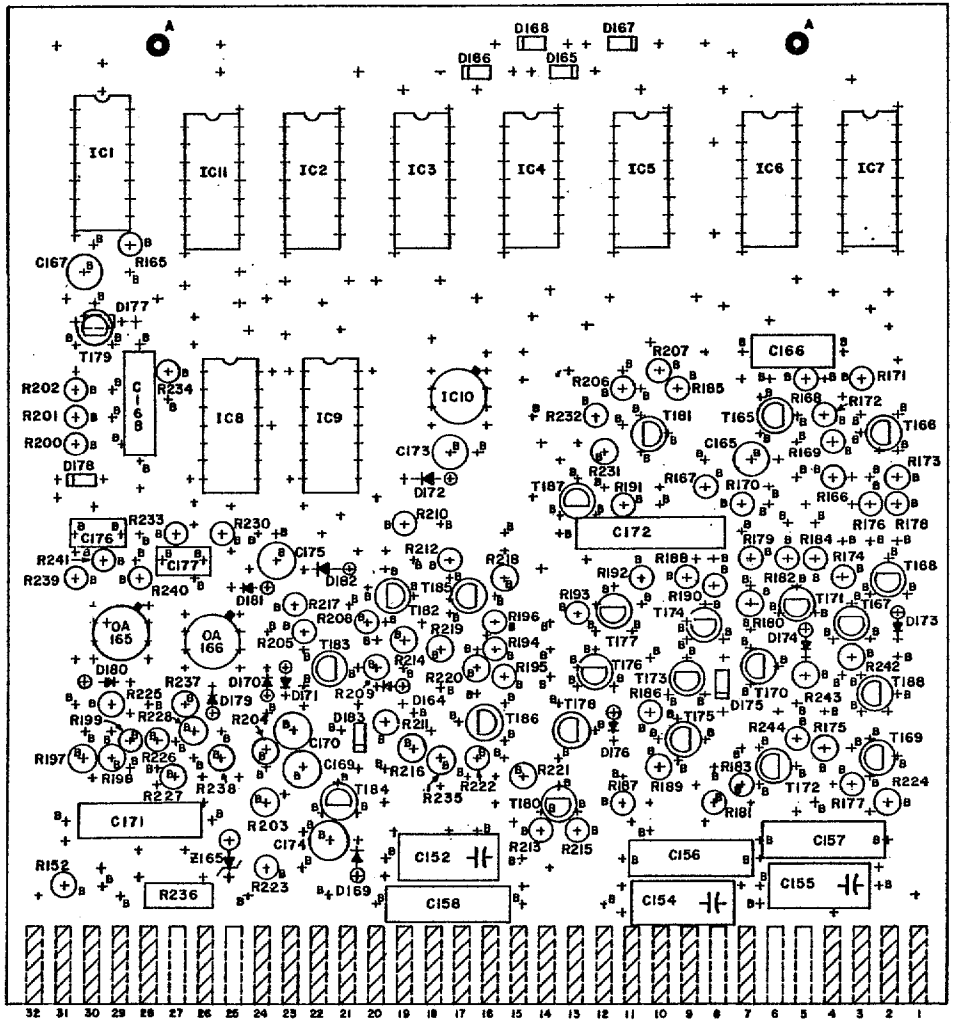
HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.040	269

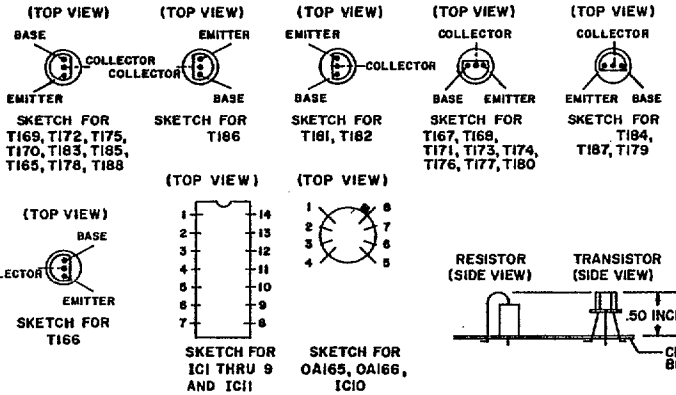
NOTES

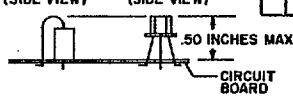
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.016}$ X $5.130^{+.002}_{-.008}$



SEE NOTE 1 & 2

TRANSISTOR LEAD SKETCHES





2X SCALE	36A353788AB005 REAR ETCHED BOARD DWG.	36A353789AB005 FRONT ETCHED BOARD DWG.	1 FRONT SPACE UNIT REQUIREMENT	0 BACK

REVISIONS

REV NO. 0

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

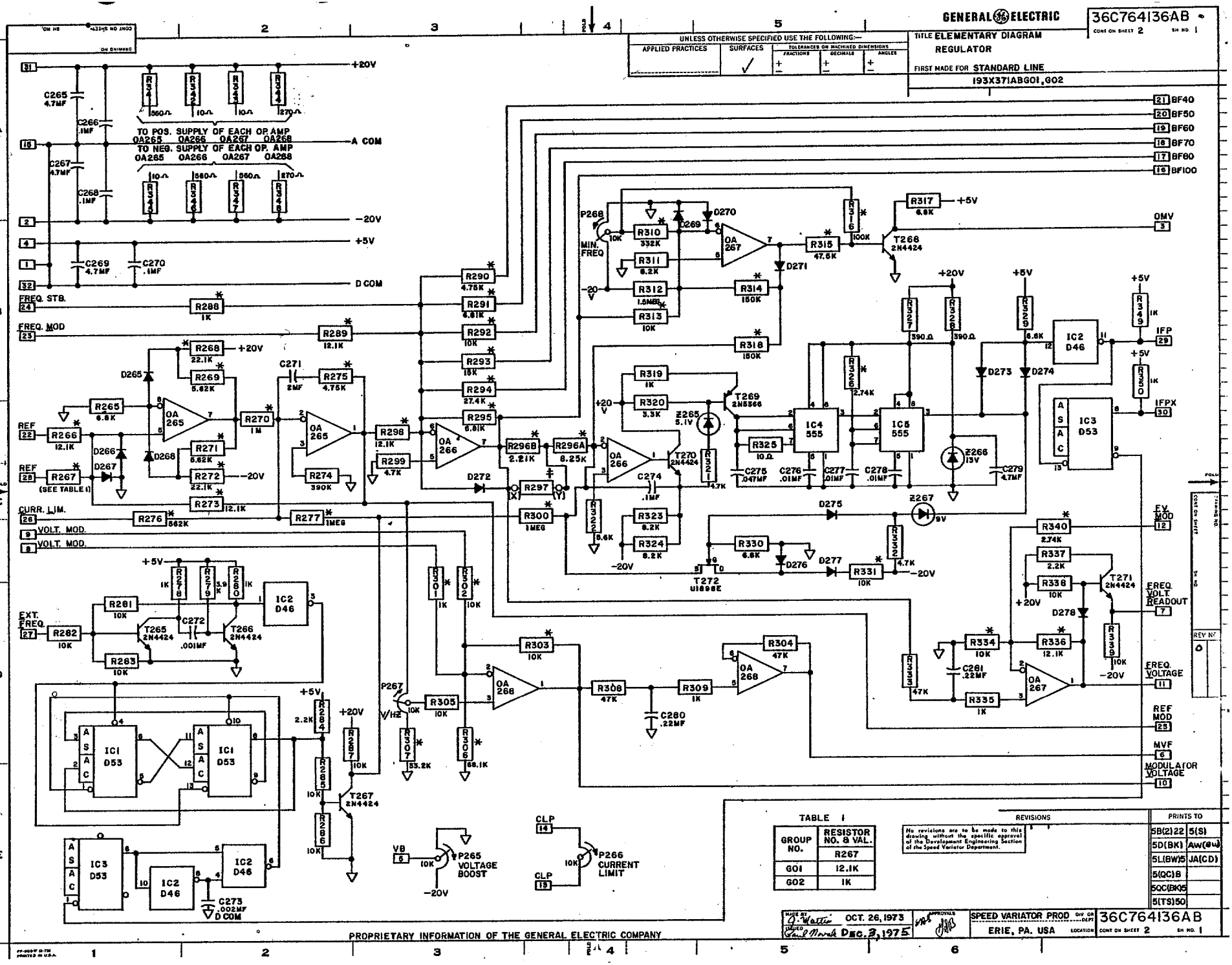
5B(2)22	
5D(BK)	JA(CD)
5(QC)8	5L(BW)5
5QC(BK)5	5AE(BW)
5(TS)50	
5(S)	

DRAWN BY J.R. Baker MAY 25, 1977
ISSUED R. Brewer MAY 25, 1977
APPROVALS

SPEED VARIATOR PROD DEPT OR DEPT
ERIE, PA. LOCATION

36C764126AC
CONT ON SHEET FL. SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764136AB
TITLE ELEMENTARY DIAGRAM
REGULATOR
FIRST MADE FOR STANDARD LINE
193X371ABG01, G02
+20V
-20V
+5V
A COM
D COM
TO POS. SUPPLY OF EACH OP. AMP
OA265 OA266 OA267 OA268
TO NEG. SUPPLY OF EACH OP. AMP
OA265 OA266 OA267 OA268
FREQ. STB.
FREQ. MOD
REF
CURR. LIM.
VOLT. MOD.
EXT. FREQ.
MIN. FREQ
V/Hz
VB
P265 VOLTAGE BOOST
P266 CURRENT LIMIT
CLP
IFP
IFPX
EV MOD
FREQ. VOLT READOUT
FREQ. VOLTAGE
REF MOD
MVF
MODULATOR VOLTAGE
OMV
BF40
BF50
BF60
BF70
BF80
BF100
TABLE I
GROUP NO.
RESISTOR NO. & VAL.
R267
G01
12.1K
G02
1K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(2)22 5(S)
5D(BK) AW(GW)
5L(BW)5 JA(CD)
5(QC)8
5QC(BK)5
5(TS)50
OCT. 26, 1973
Dec. 3, 1975
SPEED VARIATOR PROD
ERIE, PA. USA
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

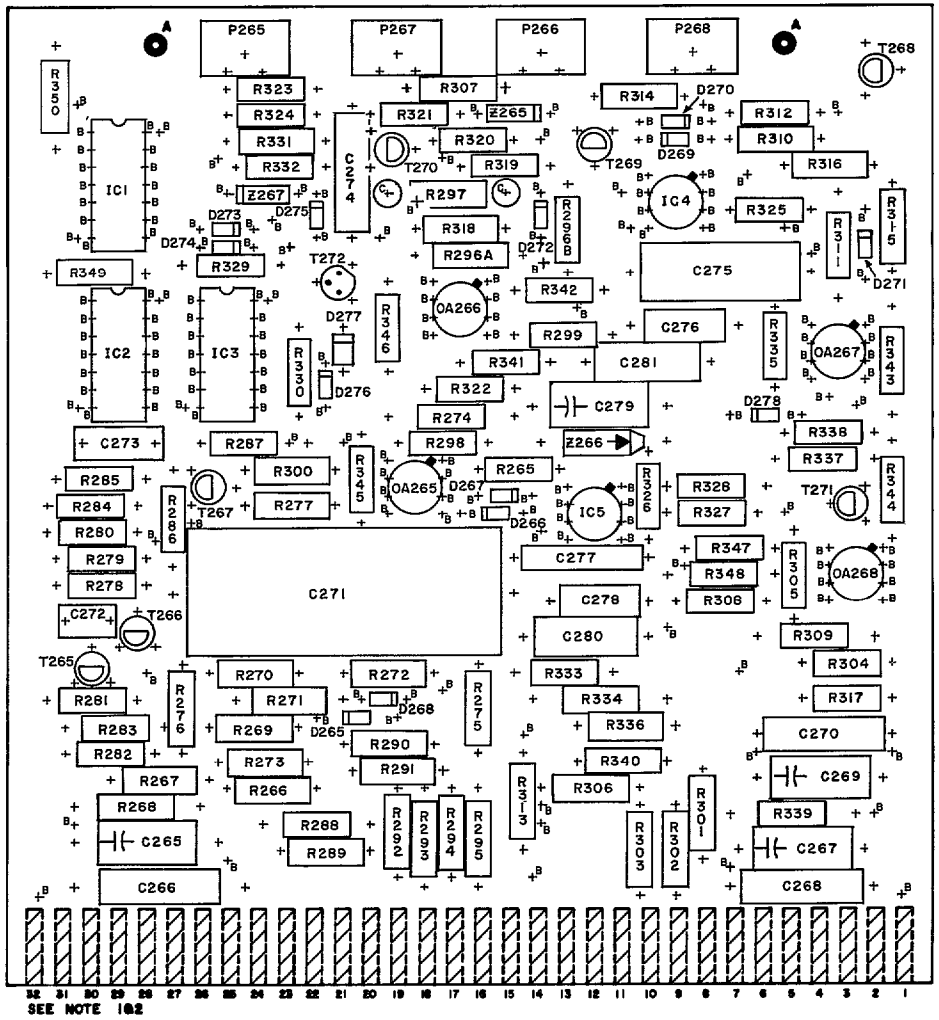
GENERAL ELECTRIC

36C764136AB
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES OR MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
710A815	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
REGULATOR CARD
FIRST MADE FOR STANDARD LINE AF-3061
193X371ABG01,G02



HOLE TABULATION

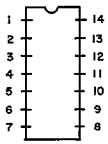
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN
A	.157	2
B	032	168
C	081	2

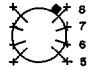
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.
5. FOR-G02 ONLY
   STAMP ADHESIVE LABEL-G02. ADD LABEL AS SUFFIX TO 193X371AA IN PLACE OF G01 STAMPED ON PRINTED CIRCUIT BOARD.

IC & OA LEAD SKETCHES

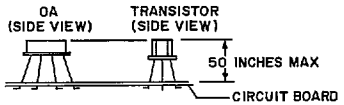


SKETCH FOR IC1, IC2, IC3

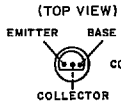


SKETCH FOR OA265, OA266, OA267, OA268 IC4, IC5

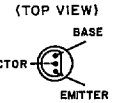
COMPONENT HEIGHT SKETCH



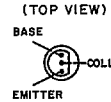
TRANSISTOR LEAD SKETCHES



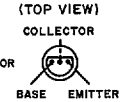
SKETCH FOR T269



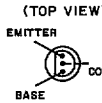
SKETCH FOR T271



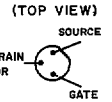
SKETCH FOR T268, T267



SKETCH FOR T265, T266



SKETCH FOR T270



SKETCH FOR T272

2X	36A353697AC005	36A353698AC005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

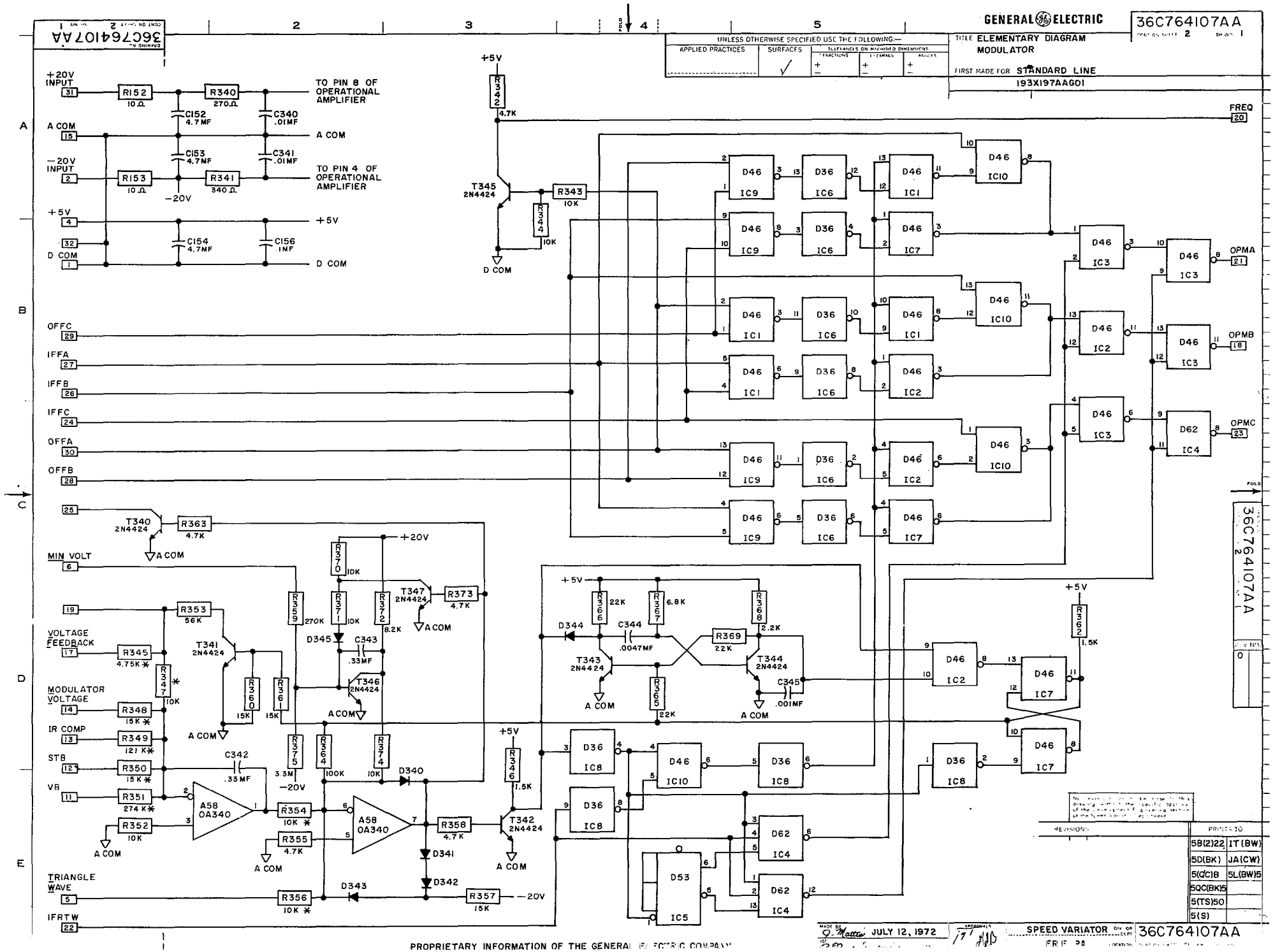
1	M Cuddy Nov 30, 1978	PER' SV-893#37	

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

DRAWN BY	DATE	APPROVALS			
G. Mattes	NOV 21, 1973	ERS	SPEED VARIATOR PROD DIV. OR DEPT	36C764136AB	
M Cuddy	Nov. 30, 1978		ERIE, PA USA LOCATION	CONT ON SHEET FL	SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764107AA
TITLE ELEMENTARY DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01
+20V INPUT
-20V INPUT
A COM
D COM
+5V
TO PIN 8 OF OPERATIONAL AMPLIFIER
TO PIN 4 OF OPERATIONAL AMPLIFIER
R152
R153
R340
R341
C152
C153
C154
C156
C340
C341
FREQ
OPMA
OPMB
OPMC
OFFC
IFFA
IFFB
IFFC
OFFA
OFFB
MIN VOLT
VOLTAGE FEEDBACK
MODULATOR VOLTAGE
IR COMP
STB
VB
TRIANGLE WAVE
IFRTW
T340
T341
T342
T343
T344
T345
T346
T347
2N4424
A58 0A340
D53
D62
D46
D36
SPEED VARIATOR
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764107AA
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

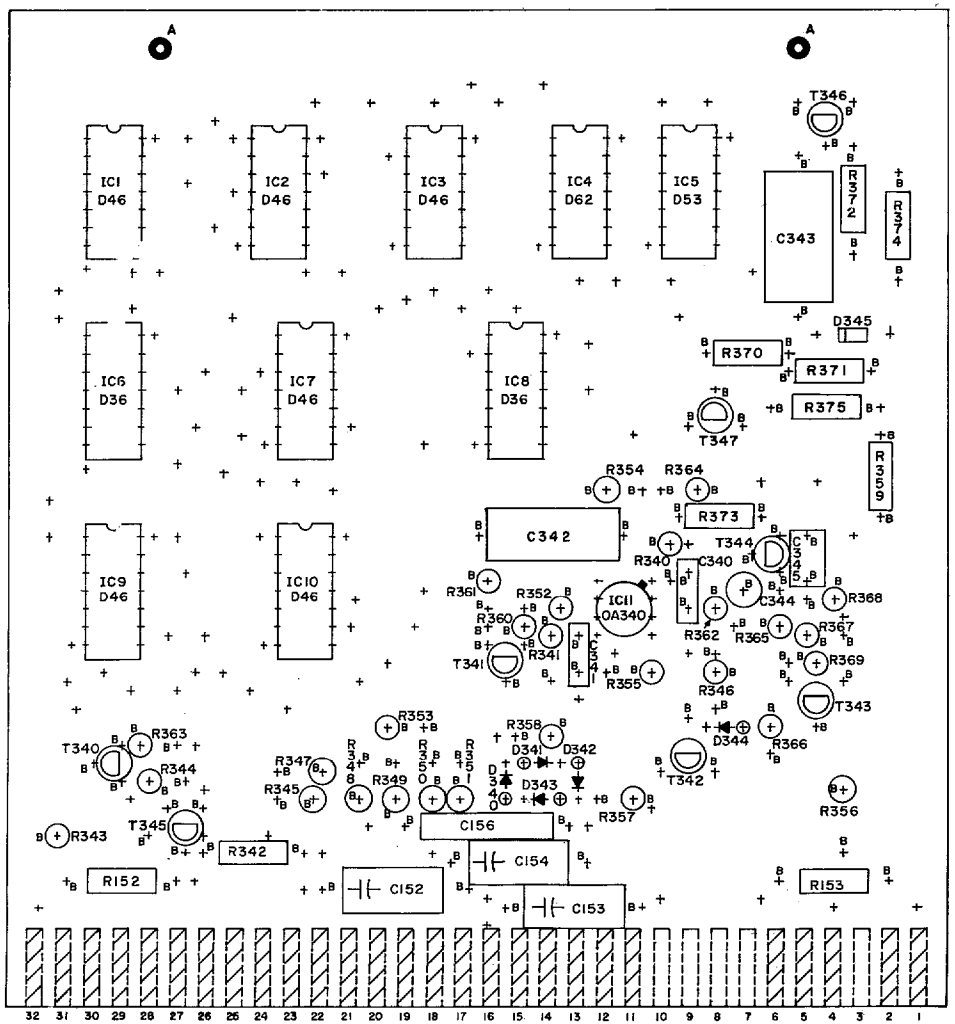
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

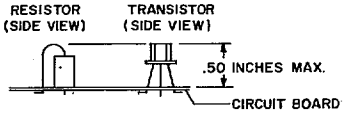
LOC.	DIA.	QUAN.
A	.157	2
B	.040	120



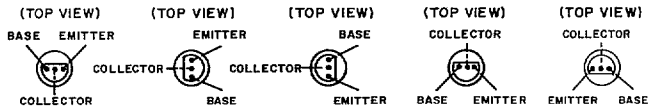
SEE NOTE 1&2

NOTES

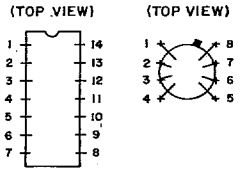
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.



TRANSISTOR LEAD SKETCHES



IC LEAD SKETCHES



2X	36A352956AA005	36A352955AA005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE	BACK UNIT REQUIREMENT

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

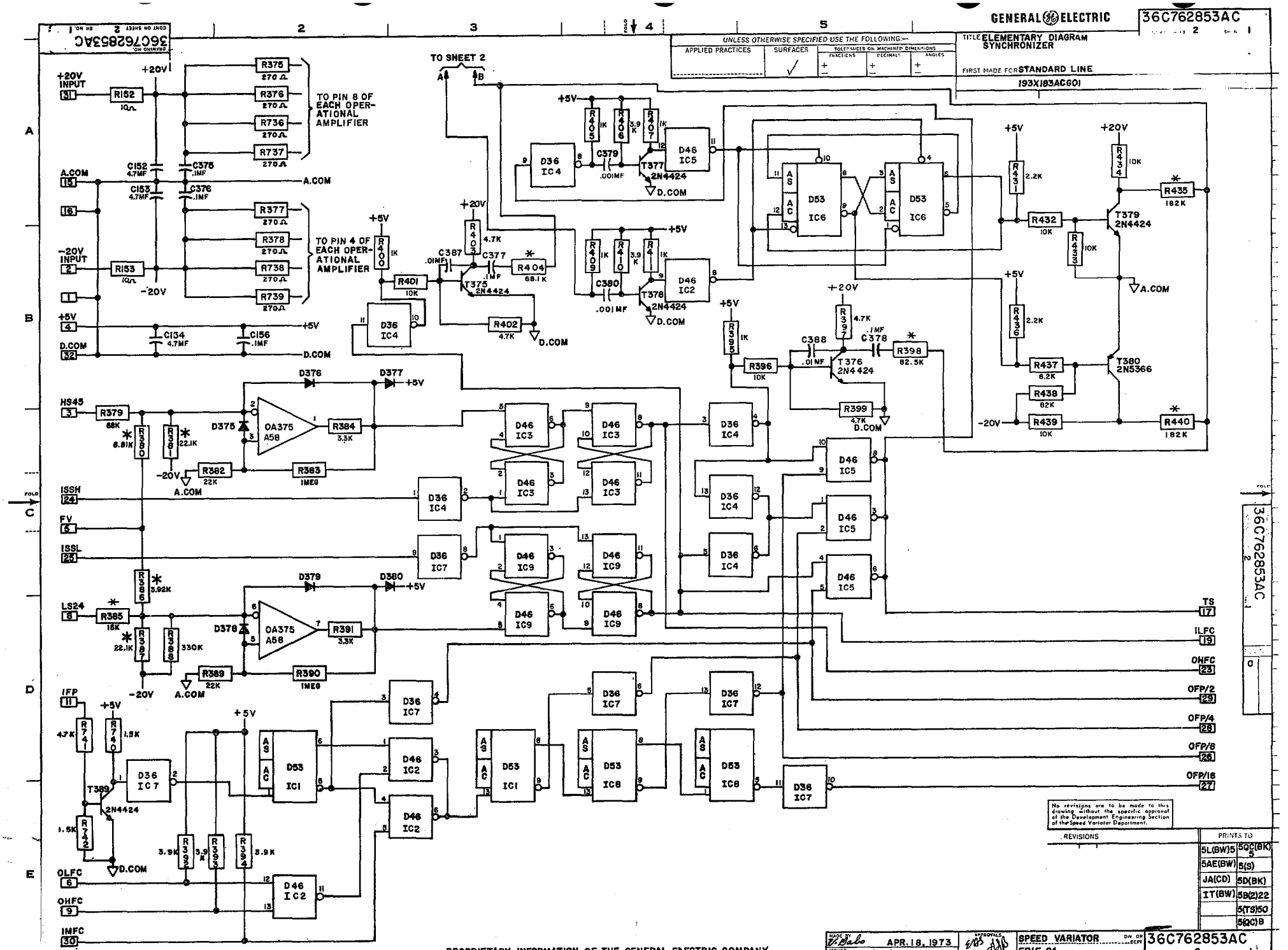
REV NO.
0

PRINTS TO	
5B(2)22	IT(BW)
5D(BK)	JA(CD)
5(QC)B	5L(BW)5
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY G. Matte AUG. 25, 1972 APPROVALS

SPEED VARIATOR DIV. OR DEPT. ERIE, PA.

36C764107AA

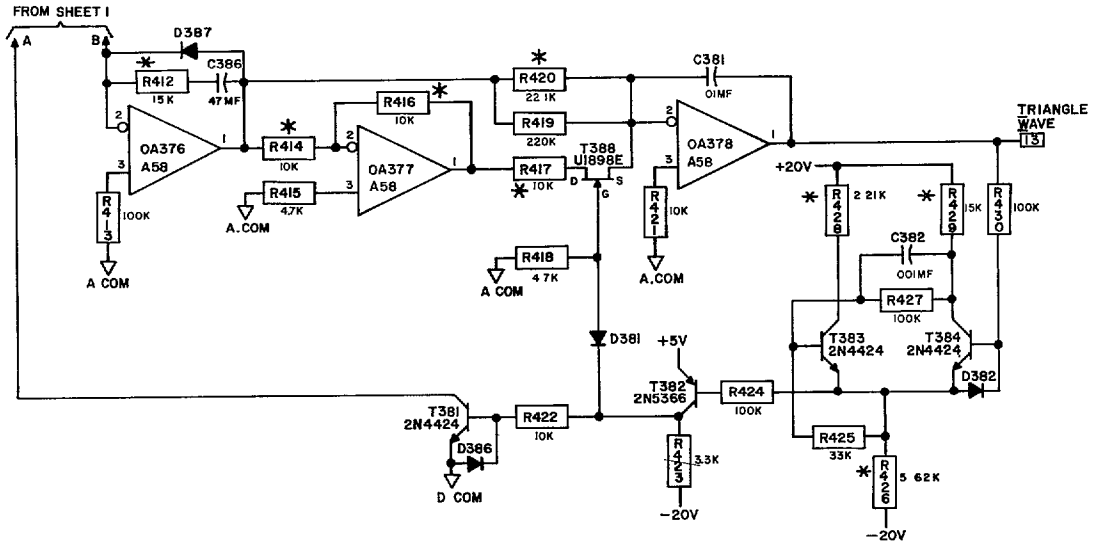


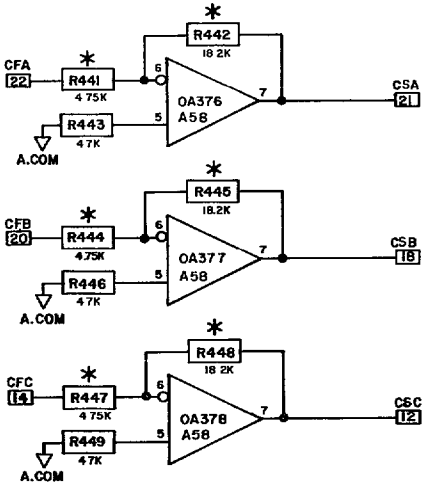
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

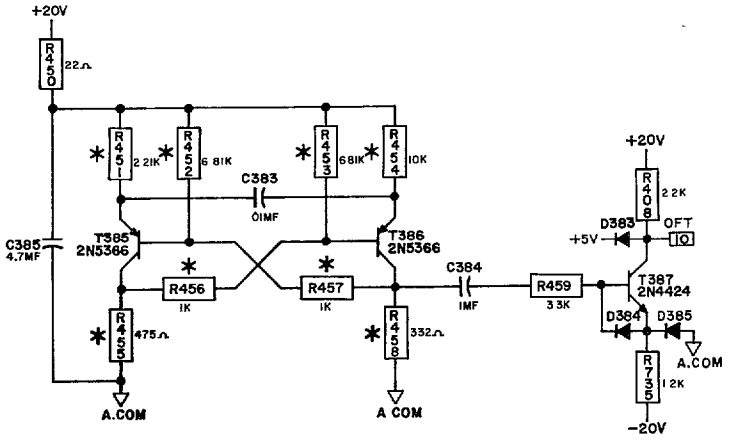
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
SYNCHRONIZER

FIRST MADE FOR STANDARD LINE
193X183ACG01







No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS

5L(BW)5	5QC(BK)
5AE(BW)	5(S)
JA(CD)	5D(BK)
IT(BW)	5B(2)22
	5(TS)50
	5(QC)8

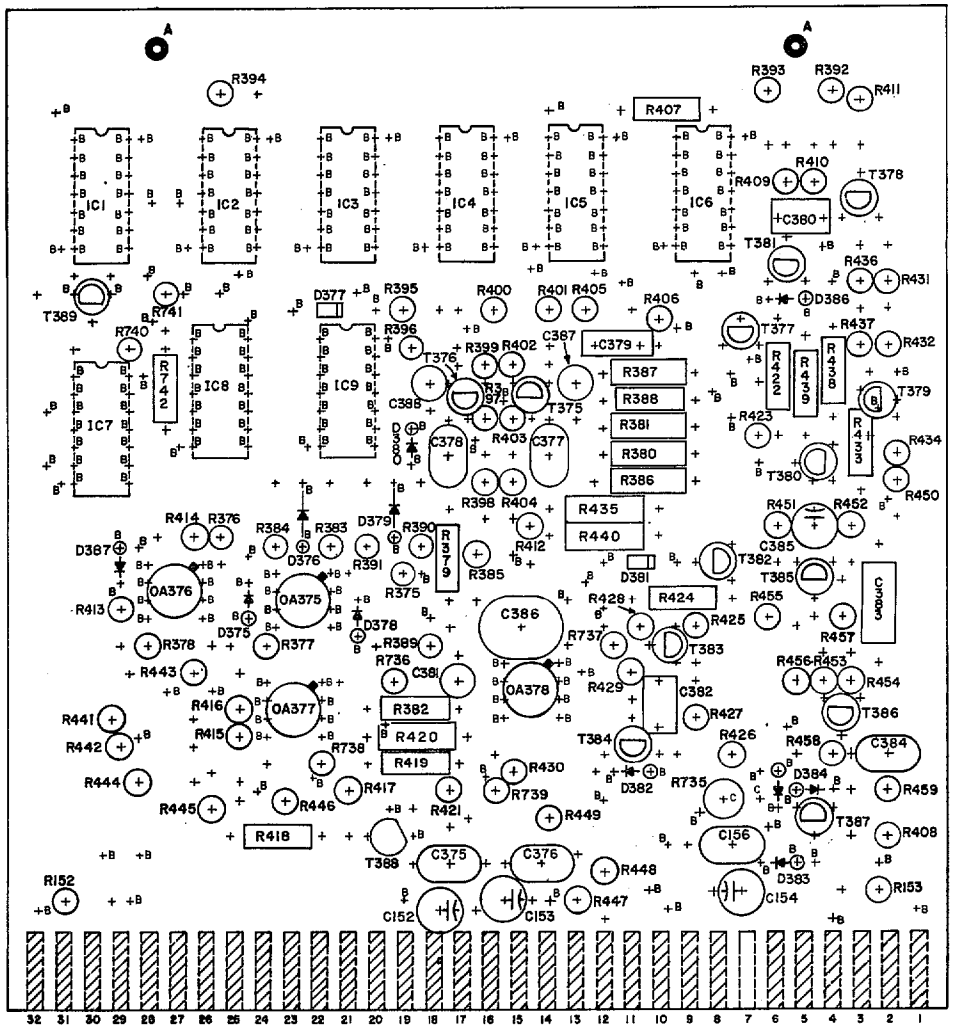
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN BY V. Sabo APR. 18, 1973
ISSUED Paul Novak MARCH 5 1974
APPROVALS
SPEED VARIATOR ERIE PA
36C762853AC
LOCATION CLT ON SHEET 3 SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING
APPLIED PRACTICES: 718A515 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
SYNCHRONIZER

FIRST MADE FOR STANDARD LINE
193X183ACG01



HOLE TABULATION

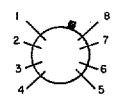
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.032	272
C	.052	2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

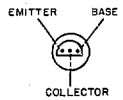
(TOP VIEW)



SKETCH FOR OA375, OA376, OA377, OA378

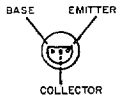
TRANSISTOR LEAD SKETCHES

(TOP VIEW)



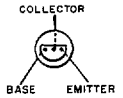
SKETCH FOR T375, T385, T387, T389

(TOP VIEW)



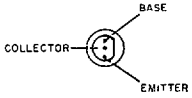
SKETCH FOR T376, T386

(TOP VIEW)



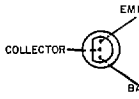
SKETCH FOR T377, T378, T381, T384

(TOP VIEW)



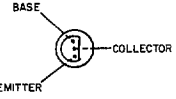
SKETCH FOR T380

(TOP VIEW)



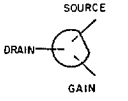
SKETCH FOR T382, T383

(TOP VIEW)



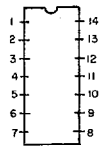
SKETCH FOR T379

(TOP VIEW)



SKETCH FOR T388

(TOP VIEW)



SKETCH FOR IC PACKAGES

2X	36A352972AC005	36A352973AC005	2	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

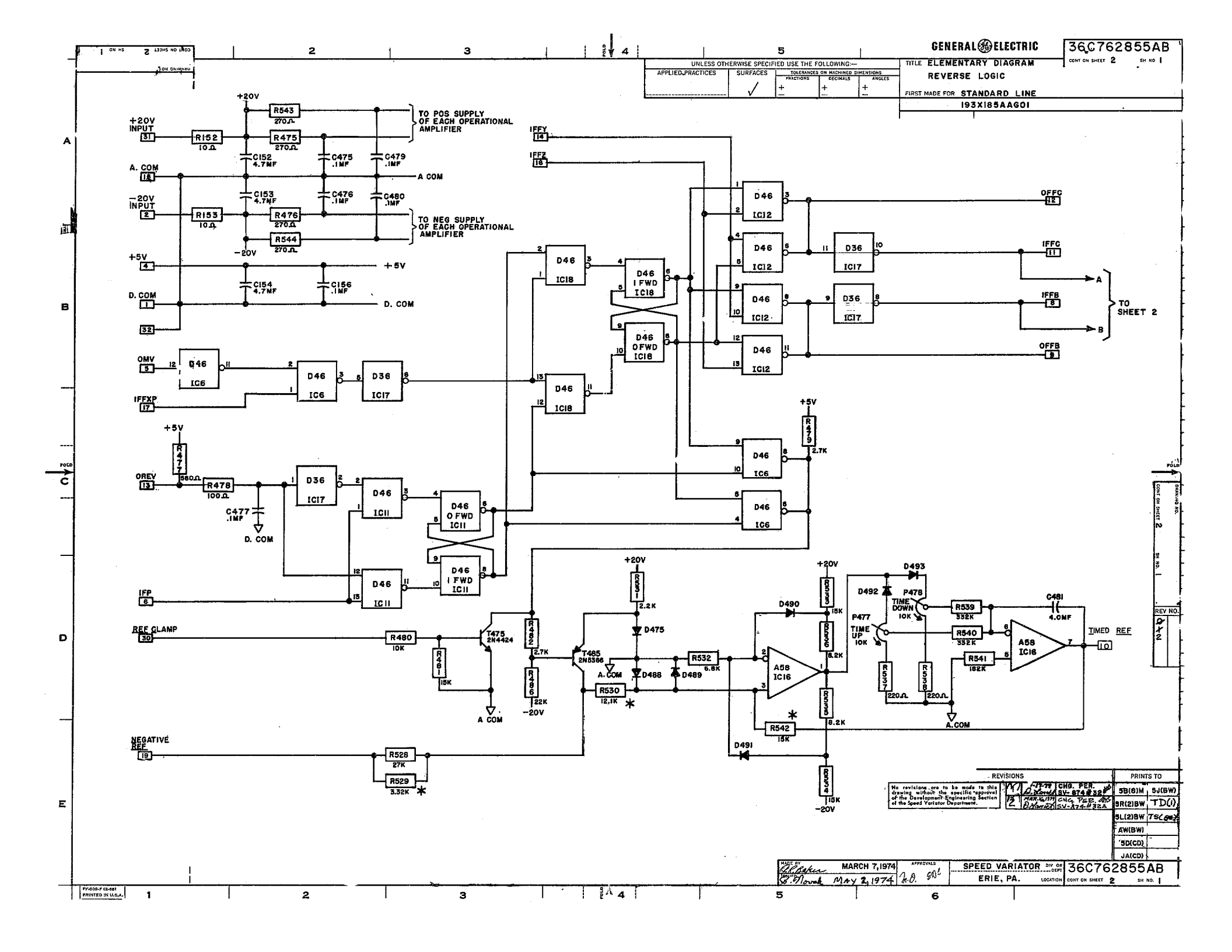
REV NO. 0

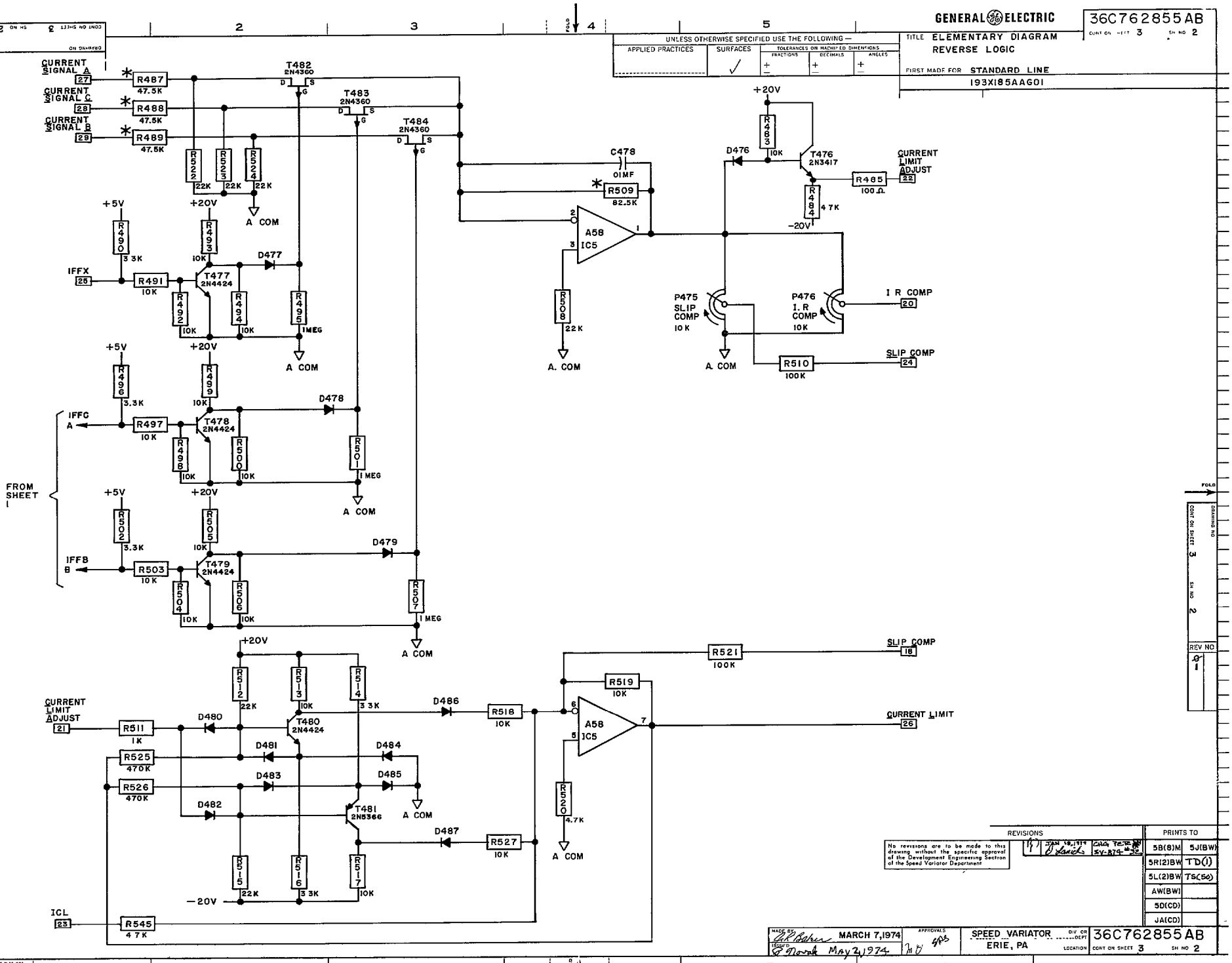
PRINTS TO	
5QC(S)	5(S)
5AE(BW)	5D(BK)
JA(CD)	5B(2)
IT(BW)	5T8(5D)
5L(BW)5	5QC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY J. Mattis MAY 18, 1973
ISSUED C. Baker MARCH 15, 1974
APPROVALS

SPEED VARIATOR
ERIE


GENERAL ELECTRIC
36C762855AB
CONT ON SHEET 2 SH NO 1
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC
FIRST MADE FOR STANDARD LINE
193X185AAG01
+20V INPUT
-20V INPUT
A. COM
D. COM
+5V
TO POS SUPPLY OF EACH OPERATIONAL AMPLIFIER
TO NEG SUPPLY OF EACH OPERATIONAL AMPLIFIER
R543 270Ω
R475 270Ω
R544 270Ω
R476 270Ω
R152 10Ω
R153 10Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C475 .1MF
C476 .1MF
C156 .1MF
C479 .1MF
C480 .1MF
A COM
D. COM
+5V
-20V
IFFY
IFFZ
OFFC
IFFC
IFFB
OFFB
TO SHEET 2
D46 IC12
D36 IC17
D46 IC18
D46 IFWD IC18
D46 OFWD IC18
OMV
IFFXP
D46 IC6
D36 IC17
OREV
R477 560Ω
R478 100Ω
C477 .1MF
D46 IC11
D46 OFWD IC11
D46 IFWD IC11
IFP
R479 2.7K
REF CLAMP
R480 10K
R481 15K
T475 2N4424
R482 2.7K
R483 22K
T485 2N5386
R530 12.1K
R531 2.2K
D475
D488
D489
R532 6.8K
D490
R533 15K
R534 8.2K
R535 8.2K
R542 15K
D491
R536 15K
A58 IC16
D492
D493
P478 TIME DOWN 10K
P477 TIME UP 10K
R539 332K
R540 332K
R541 182K
R537 220Ω
R538 220Ω
C481 4.0MF
TIMED REF
NEGATIVE REF
R528 27K
R529 3.32K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
CHG. PER.
PRINTS TO
5B(6)M
5J(BW)
9R(2)BW
TD(I)
5L(2)BW
AW(BW)
5D(CD)
JA(CD)
MADE BY
MARCH 7, 1974
APPROVALS
May 2, 1974
SPEED VARIATOR
ERIE, PA.
DIV OR DEPT
LOCATION
36C762855AB
CONT ON SHEET 2 SH NO 1
PRINTED IN U.S.A.


GENERAL ELECTRIC
36C762855AB
TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC
FIRST MADE FOR STANDARD LINE
193X185AAG01
CURRENT SIGNAL A
CURRENT SIGNAL C
CURRENT SIGNAL B
R487 47.5K
R488 47.5K
R489 47.5K
T482 2N4360
T483 2N4360
T484 2N4360
R522 22K
R523 22K
R524 22K
A COM
+5V
+20V
R490 3.3K
R493 10K
IFFX
R491 10K
T477 2N4424
D477
R492 10K
R494 10K
R495 1MEG
R496 3.3K
R499 10K
IFFC
R497 10K
T478 2N4424
D478
R498 10K
R500 10K
R501 1 MEG
FROM SHEET 1
R502 3.3K
R505 10K
IFFB
R503 10K
T479 2N4424
D479
R504 10K
R506 10K
R507 1MEG
C478 .01MF
R509 82.5K
A58 IC5
R508 22K
A. COM
D476
T476 2N3417
R483 10K
R484 4.7K
-20V
CURRENT LIMIT ADJUST
R485 100Ω
P475 SLIP COMP 10K
P476 I.R COMP 10K
I R COMP
SLIP COMP
R510 100K
R521 100K
R519 10K
R512 22K
R513 10K
R514 3.3K
CURRENT LIMIT ADJUST
R511 1K
D480
T480 2N4424
D486
R518 10K
D481
D484
R525 470K
D483
D485
R526 470K
D482
T481 2N5366
D487
R527 10K
R515 22K
R516 3.3K
R517 10K
R520 4.7K
CURRENT LIMIT
ICL
R545 4.7K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department
PRINTS TO
5B(8)M 5J(BW)
5R(2)BW TD(1)
5L(2)BW TS(56)
AW(BW)
5D(CD)
JA(CD)
MADE BY MARCH 7,1974
ISSUED May 2,1974
SPEED VARIATOR
ERIE, PA
36C762855AB
CONT ON SHEET 3
SH NO 2
PRINTED IN U.S.A.

GENERAL ELECTRIC

36C762855AB
CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
7168515	✓	±	±	±

TITLE **PRINTED CIRCUIT DIAGRAM**
REVERSE LOGIC

FIRST MADE FOR STANDARD LINE

193X185AAG01

HOLE TABULATION

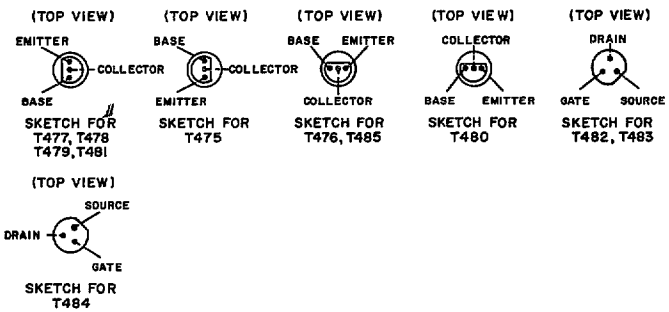
ALL HOLES 032 DIA. EXCEPT THE HOLES TABULATED BELOW

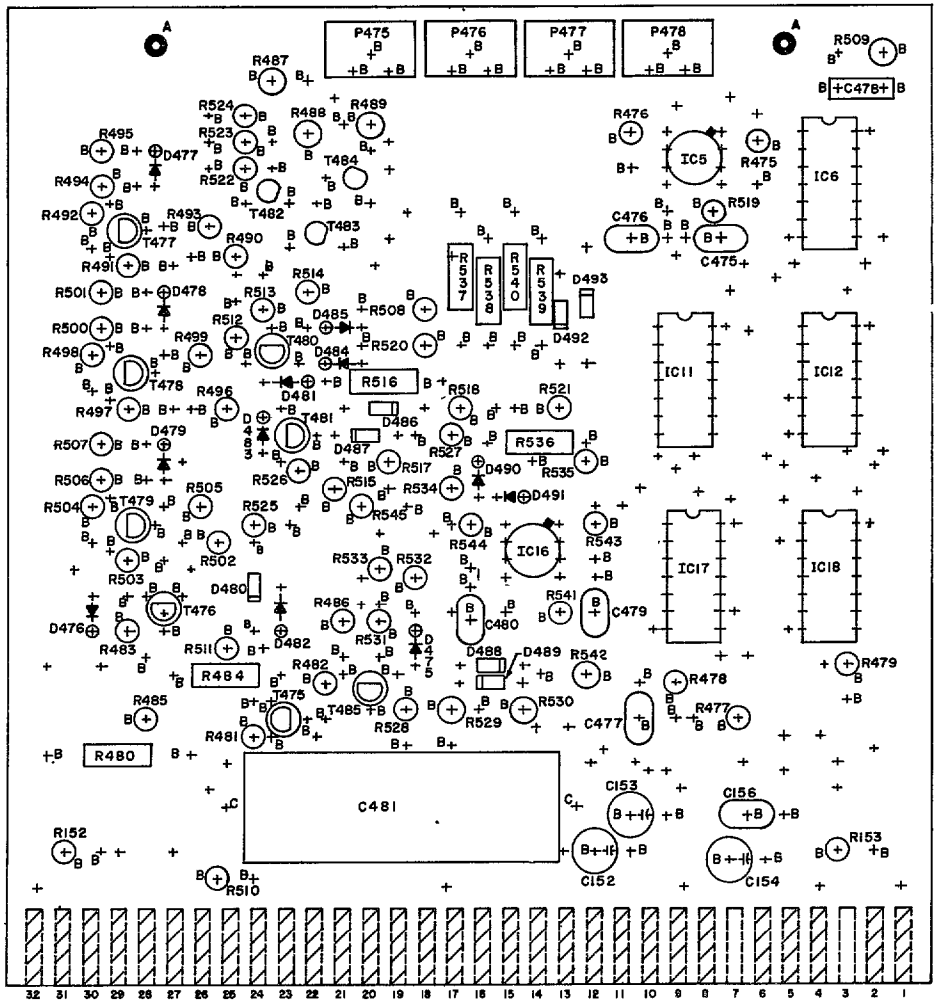
LOC.	DIA.	QUAN
A	.157	2
B	.040	213
C	.052	2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES





2X	36A352970AB005	36A352971AB005	2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT	BACK
			SPACE UNIT REQUIREMENT	

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REV NO			
1	D. Baker	July 8, 1974	CHG PER ECO SV-874-452
2	D. Baker	Nov 14, 1974	CHG PER ECO SV-874-521
3	D. Novak	Nov 16, 1979	CHG PER ECO SV-874-521

PRINTS TO

5B(8)M	5W(BW)
5R(2)BW	TDC(1)
5L(2)BW	TSC(1)
5W(BW)	
5D(CD)	
5A(CD)	

MADE BY E. Baker MARCH 8, 1974
ISSUED G. Novak May 2, 1974
APPROVALS M & ERS

SPEED VARIATOR — DIV OR DEPT
ERIE, PA — LOCATION

36C762855AB
CONT ON SHEET FL SH NO 3

SCHEMATIC DIAGRAM

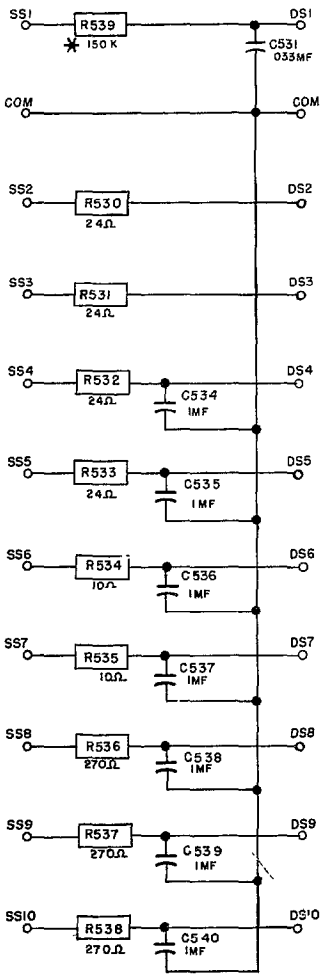
GENERAL ELECTRIC

36C762868AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

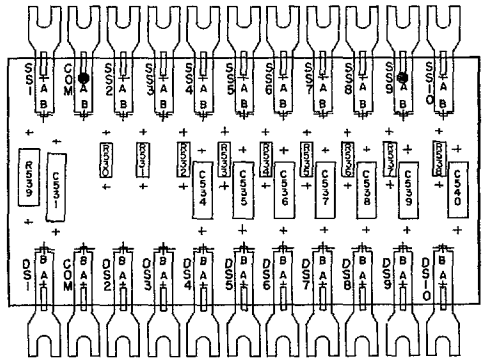
APPLIED PRACTICES	SURFACES		
718A515	✓		

PRINTED CIRCUIT DIAGRAM
SIGNAL FILTER
STANDARD LINE
193X189AAG01



✱ — INDICATES 1% RESISTANCES TOLERANCE

FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	.157	22
B	.101	22

NOTE. 1. CARD SIZE, $2.85^{+.020}$ X $5.130^{+.002}_{-.008}$

36C762868AA

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	9428522AA004		

REV

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

1	Schmitz 1-03 Nov 22, 1971	KWP	5QC(BK)55 (S) 5D(BK) 5(PCM) 50 5B(2) 5(TS)50 5(QC)B

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

R. Schmitz June 2, 1970	EAS	SPEED VARIATOR ERIE, PA.	36C762868AA
D. Schmitz JAN 27, 1971	KWP		

SCHEMATIC DIAGRAM

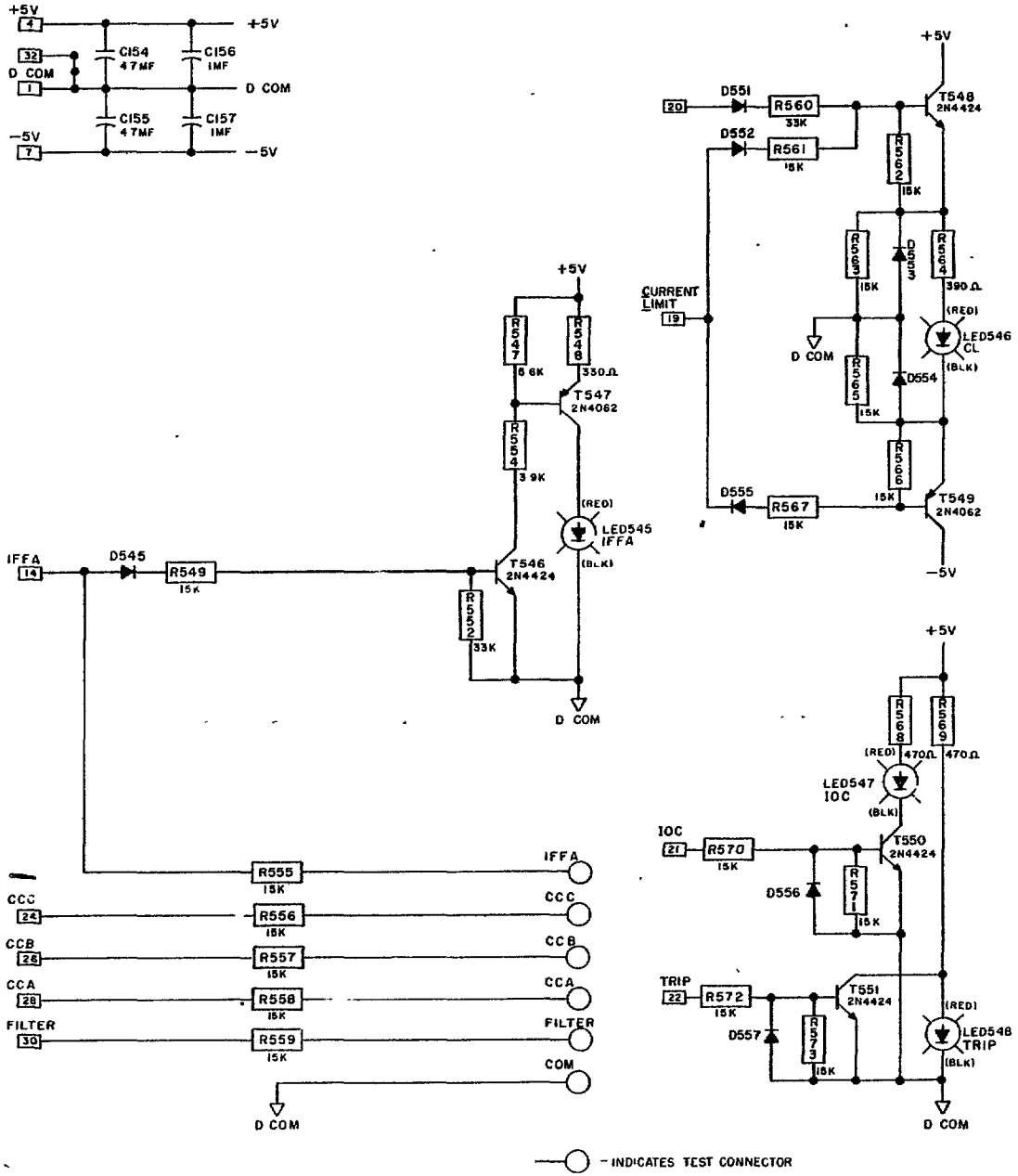
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING –				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMAL	ANGLES
718A516	✓	±	±	±

GENERAL ELECTRIC — 36C762854AA

TITLE: PRINTED CIRCUIT DIAGRAM MONITOR

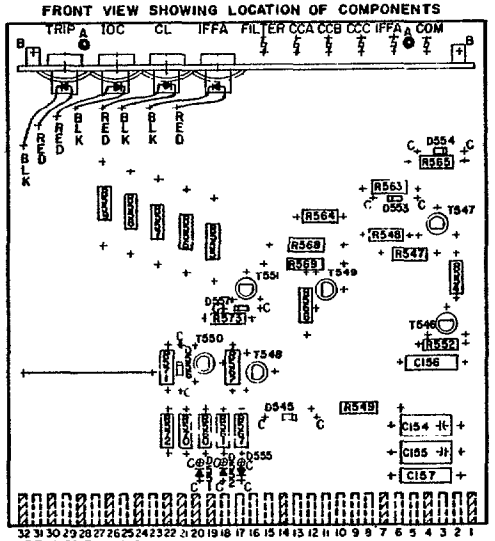
FIRST MADE FOR STANDARD LINE

193X184AAG01



○ – INDICATES TEST CONNECTOR

FRONT VIEW SHOWING LOCATION OF COMPONENTS



SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA.	QUAN
A	.157	2
B	.078	2
C	.032	16

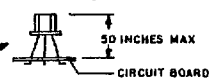
NOTE 1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

2 CROSS HATCHED TABS INDICATE TABS USED

3 CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

4 ALL TRANSISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN HERE

(SIDE VIEW)



TRANSISTOR LEAD SKETCHES



36C762854AA

REV NO: 0

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428526AA004	1	0

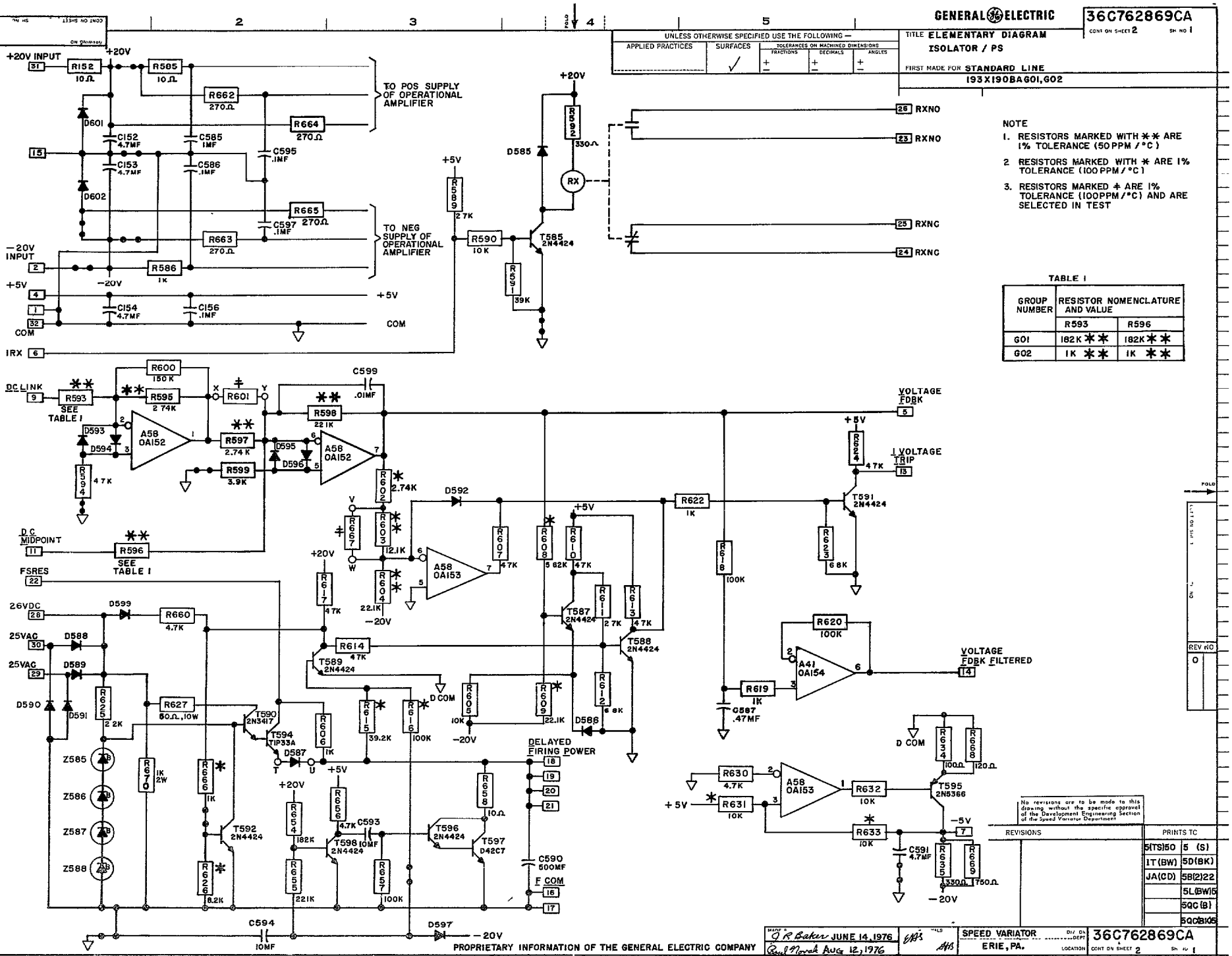
REVISIONS

PRINTED

5QC(BK)5 (S)
5D(BK)
5B(2)
5(TS)50
5(QC)B

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY	APPROVALS			
Q Britton JUNE 3, 1970	GAS	SPEED VARIATOR	CODE IDENT NO.	36C762854AA
ISSUED K Zimmer FEB 15, 1971	KKP	ERIE, PENNA.	LOCATION	SHEET


GENERAL ELECTRIC
36C762869CA
CONT ON SHEET 2
SH NO 1
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
ISOLATOR / PS
FIRST MADE FOR STANDARD LINE
193X190BAG01,G02
+20V INPUT
-20V INPUT
+5V
COM
IRX
TO POS SUPPLY OF OPERATIONAL AMPLIFIER
TO NEG SUPPLY OF OPERATIONAL AMPLIFIER
RXNO
RXNO
RXNC
RXNC
NOTE
1. RESISTORS MARKED WITH ** ARE 1% TOLERANCE (50 PPM / °C)
2 RESISTORS MARKED WITH * ARE 1% TOLERANCE (100 PPM / °C)
3. RESISTORS MARKED + ARE 1% TOLERANCE (100PPM/°C) AND ARE SELECTED IN TEST
TABLE I
GROUP NUMBER
RESISTOR NOMENCLATURE AND VALUE
R593
R596
G01
182K **
182K **
G02
1K **
1K **
DC LINK
SEE TABLE I
DC MIDPOINT
FSRES
VOLTAGE FDBK
VOLTAGE TRIP
VOLTAGE FDBK FILTERED
26VDC
25VAC
25VAC
DELAYED FIRING POWER
D COM
F COM
Z585
Z586
Z587
Z588
T585 2N4424
T586 2N4424
T587 2N4424
T588 2N4424
T589 2N4424
T590 2N3417
T591 2N4424
T592 2N4424
T594 TIP33A
T595 2N5366
T596 2N4424
T597 D42C7
T598 2N4424
A58 0A152
A58 0A153
A41 0A154
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department
REVISIONS
PRINTS TC
5(TS)50
5 (S)
1T(BW)
5D(BK)
JA(CD)
5B(2)22
5L(BW)5
5QC(B)
5QC(BK)5
MADE J R Baker JUNE 14, 1976
Paul Novak AUG 12, 1976
SPEED VARIATOR
ERIE, PA.
DIV OR DEPT
LOCATION
36C762869CA
CONT ON SHEET 2
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
FF 809 F 12 681
PRINTED IN U.S.A.

GENERAL ELECTRIC

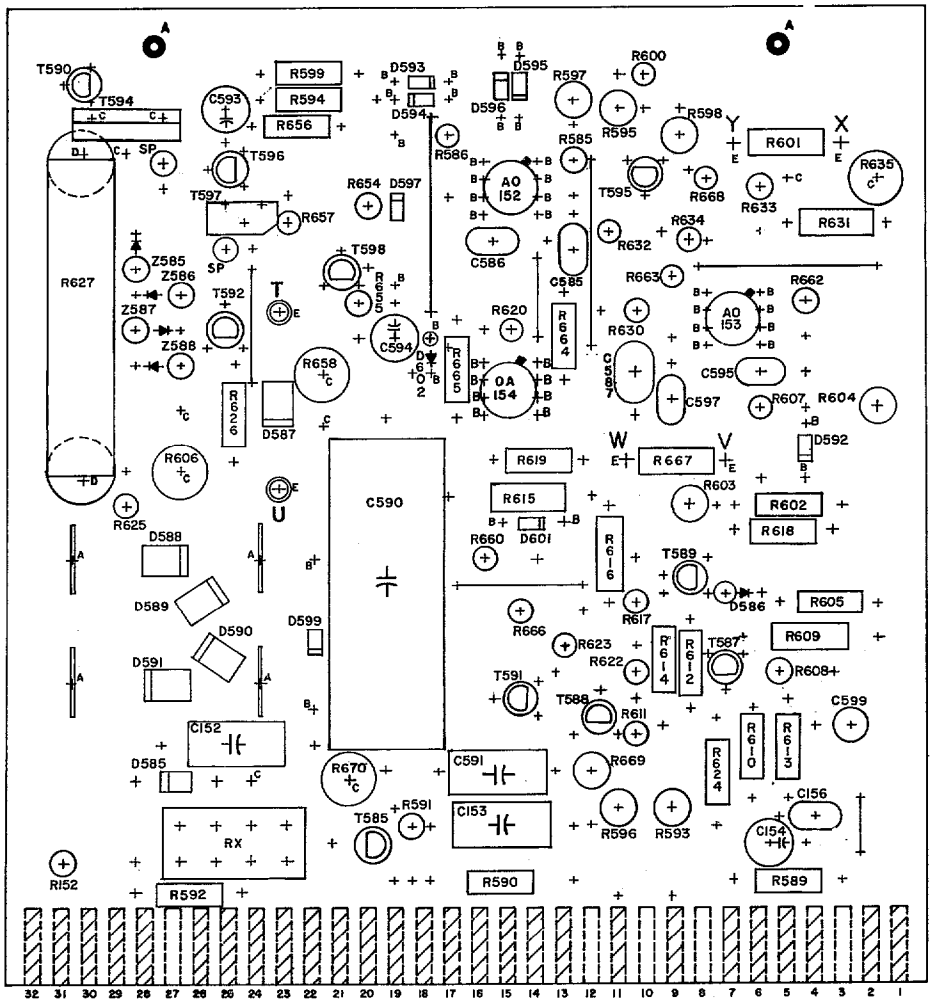
36C762869CA
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
710A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
ISOLATOR / PS

FIRST MADE FOR STANDARD LINE
193X190BAG01,G02



HOLE TABULATION

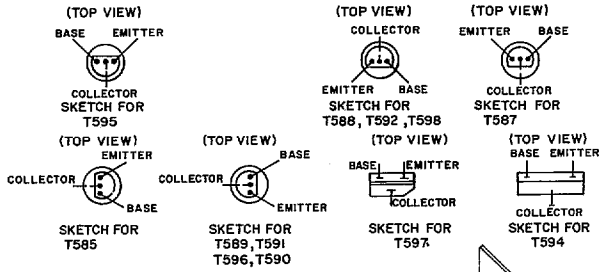
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

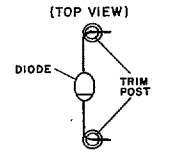
LOC.	DIA.	QUAN.
A	.157	6
B	.032	42
C	.062	11
D	.052	2
E	.078	6

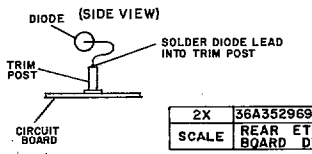
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

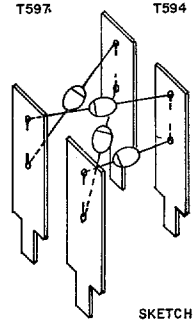
TRANSISTOR LEAD SKETCHES







SKETCH FOR MTG D587



SKETCH FOR DIODE ASM. (D588, D589, D590, D591)

2X	36A352969CA004		2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5QC(B)	5 (S)
5QC(BK) 5	5D(BK)
	5B(2)22
	5TS(BK)

REV NO. 0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY	DATE	APPROVALS			
G.R. Baker	MAR. 13, 1975	EDS	SPEED VARIATOR	DIV OR DEPT	36C762869CA
ISSUED Louis Novak	Aug. 12, 1976	HHS	ERIE	LOCATION	CONT ON SHEET FL SH NO 2

GENERAL ELECTRIC

36C762870AB

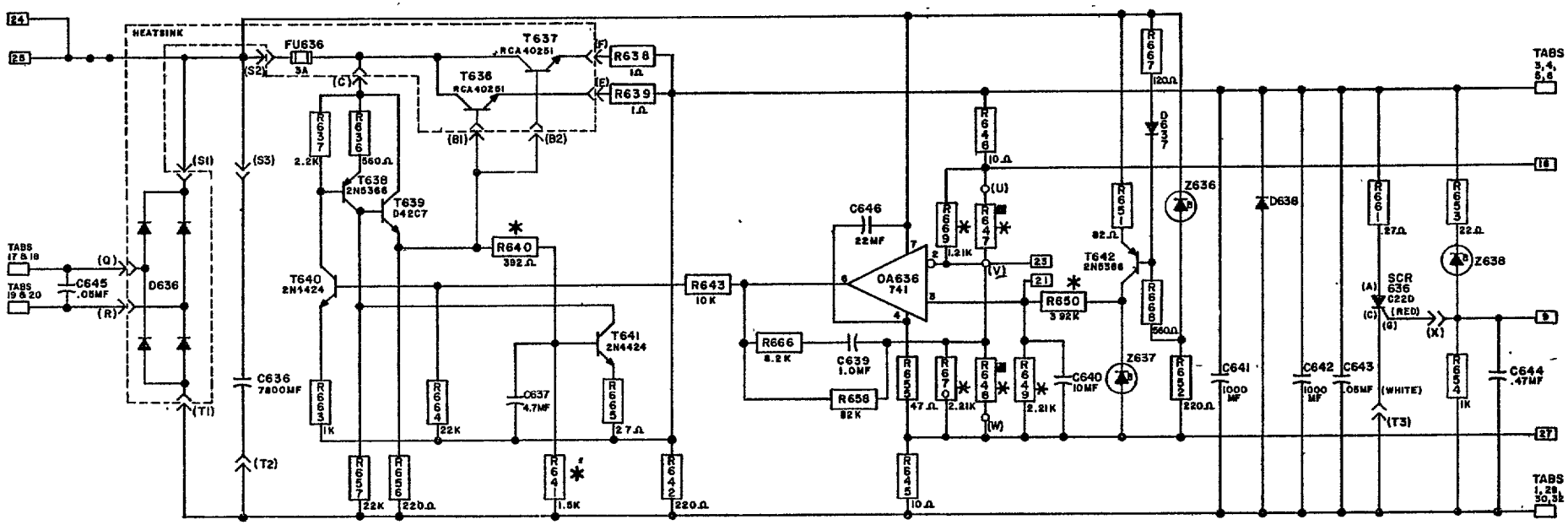
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING –

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE ELEMENTARY DIAGRAM
+5 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X191ABG01

CONT ON SHEET 2 SH NO. 1



✱ INDICATE 1% RESISTOR

■ RESISTORS SELECTED IN TEST

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
5QC(BK)	6(S)
1T(BW)	50(BK)
JA(CD)	58(2)22
5AE(BW)	5L(BW)5
	5(TS)50
	5(QC)B

REV NO. 0

CONT ON SHEET SH NO. DRAWING NO.

MADE BY O. Matte AUG. 25, 1973
REVISED D. LaRue FEB. 1, 1974

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

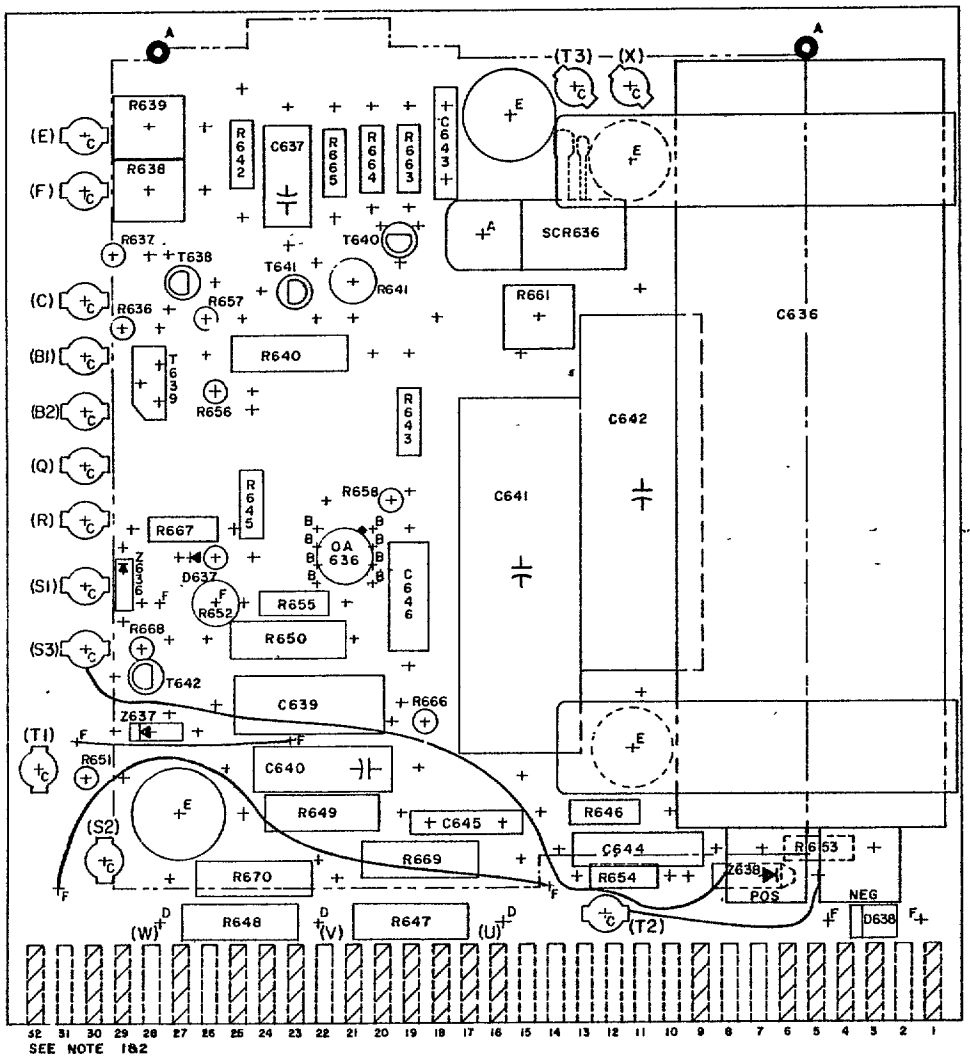
36C762870AB
CONT ON SHEET 2 SH NO. 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
+ 5 VOLT POWER SUPPLY
FIRST MADE FOR STANDARD LINE
193X191ABG01



HOLE TABULATION

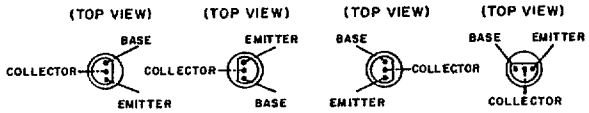
ALL HOLES .040 DIA EXCEPT THE HOLES TABULATED BELOW

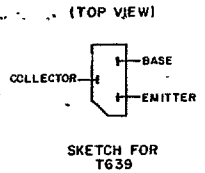
LOC.	DIA.	QUAN
A	.157	3
B	.032	8
C	.101	14
D	.078	3
E	.187	4
F	.062	8

NOTES

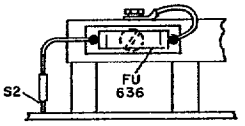
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES

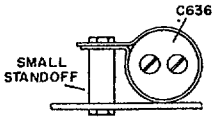




FUSE HOLDER CONNECTION SKETCH
(SIDE VIEW OF HEATSINK FROM TAB END OF CARD)



CAPACITOR MOUNTING SKETCH



LEAD CONNECTIONS

CONNECT RING TERMINAL OF LEAD S3 TO POSITIVE TERMINAL OF C636
CONNECT RING TERMINAL OF LEAD T2 TO NEGATIVE TERMINAL OF C636
COUPLE RED LEAD FEMALE CONNECTOR TO MALE CONNECTOR X AND SCR636 WHITE LEAD FEMALE CONNECTOR TO MALE CONNECTOR T3
COUPLE REMAINING LEADS WITH FEMALE CONNECTORS TO MALE CONNECTORS BEARING NOMENCLATURE IDENTICAL TO THE LEAD MARKING

REV NO
0

2X	36A352968AB004	3	0
SCALE	ETCHED CIRCUIT BOARD DWG.	FRONT SPACE	BACK UNIT REQUIREMENT

REVISIONS

PRINTS TO

5L(BW)15	5 (S)
1T(BW)	5D(BK)
JA(CD)	5B(2)22
5AE(BW)	5T5(BC)
	5QC(BK)
	5QC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY	DATE	APPROVALS		
G. Mattia	SEPT 4, 1973		SPEED VARIATOR	36C762870AB
ISSUED	FEB 1 1974		ERIE, PA	CONT ON SHEET FL SH NO 2

GENERAL ELECTRIC

36C762896AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

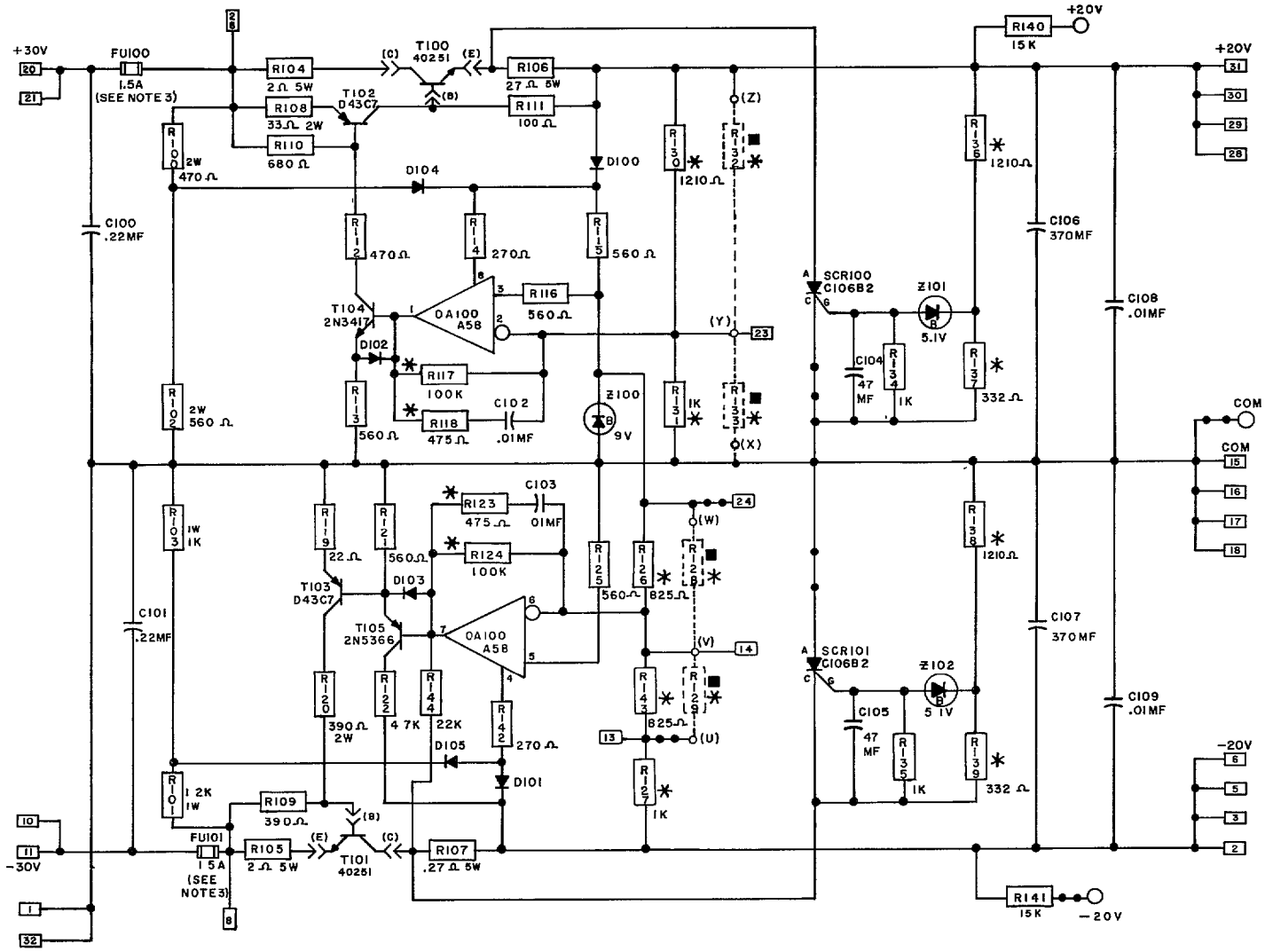
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A505	✓	±	±	±

TITLE ELEMENTARY DIAGRAM

20 VOLT POWER SUPPLY CARD

FIRST MADE FOR STANDARD LINE

193X257AAG01



NOTES 1 ✱ — 1% TEMPERATURE COMPENSATED RESISTOR.
■ — RESISTOR SELECTION IN TEST

2 NUMBERS INSIDE SMALL RECTANGLES INDICATE TAB NUMBERS WHICH CORRESPOND TO MATCHING RECEPTACLE NUMBERS

3 FUSES ARE INSTRUMENT TYPE, LITTLE FUSE 361001, OR EQUIVALENT

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS

REV NO 0

PRINTS TO	
JA(CD)	5(S)
IT(BW)	50C(BK) 5
5L(BW)5	5D(BK)
5AE(BW)	58(2)22
	5(TS)50
	5(OC)BK

MADE BY R.R. Baker MARCH 15, 1974
CHECKED E. Mount SEPT 4, 1974
APPROVALS dcg DHS
speed variator — ERIE, PA

36C762896AB CONT ON SHT 2 SH NO 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

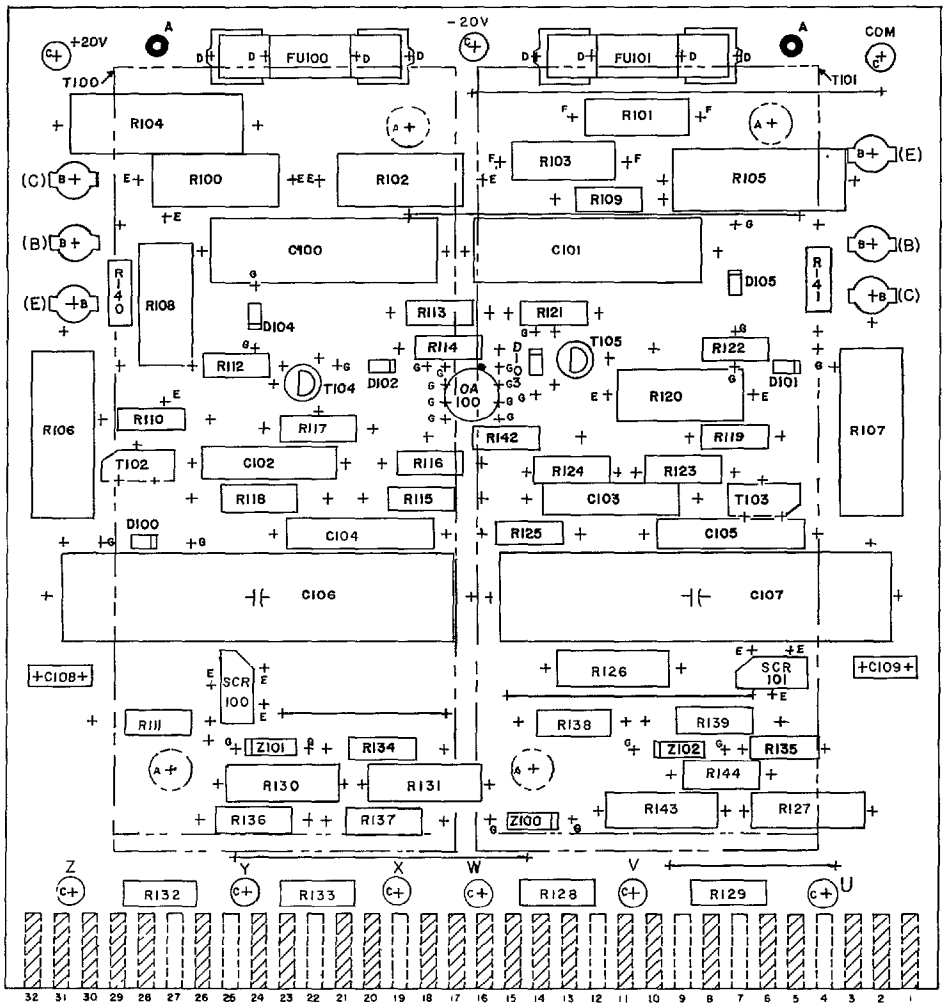
36C762896AB

PRINTED CIRCUIT DIAGRAM
20 VOLT POWER SUPPLY

FOR STANDARD LINE

193X257AA601

APPLIED PRACTICES 718A515



GROUP	KEY LOCATIONS			
01	6-7	12-13	26-27	—

HOLE TABULATION

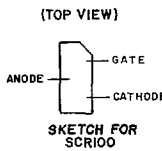
ALL HOLES .040 DIA EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	.157	6
B	.101	6
C	.081	9
D	.078	8
E	.062	14
F	.052	4
G	.032	26

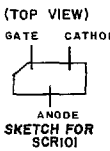
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

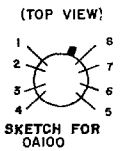
SCR LEAD SKETCHES



SKETCH FOR SCR100

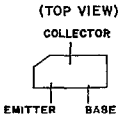


SKETCH FOR SCR101

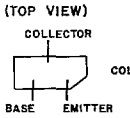


SKETCH FOR 0A100

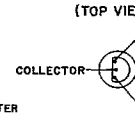
TRANSISTOR LEAD SKETCHES



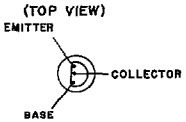
SKETCH FOR T102



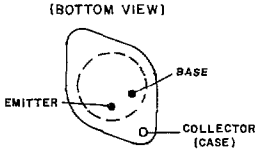
SKETCH FOR T103



SKETCH FOR T104



SKETCH FOR T105



SKETCH FOR T100, T101

2X	36A353740AB004		2	
SCALE	REAR ETCHED BOARD DWG	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

JA(CD)	5 (S)
IT(BW)	5D(BK)
5L(BW)	55B(2)22
5AE(BW)	5QC(BK)5
	5(TS)50
	5QC(B)

MARCH 15, 1974 — speed variator — ERIE, PA. — 36C762896AB FL 2

SEPT. 4 1974

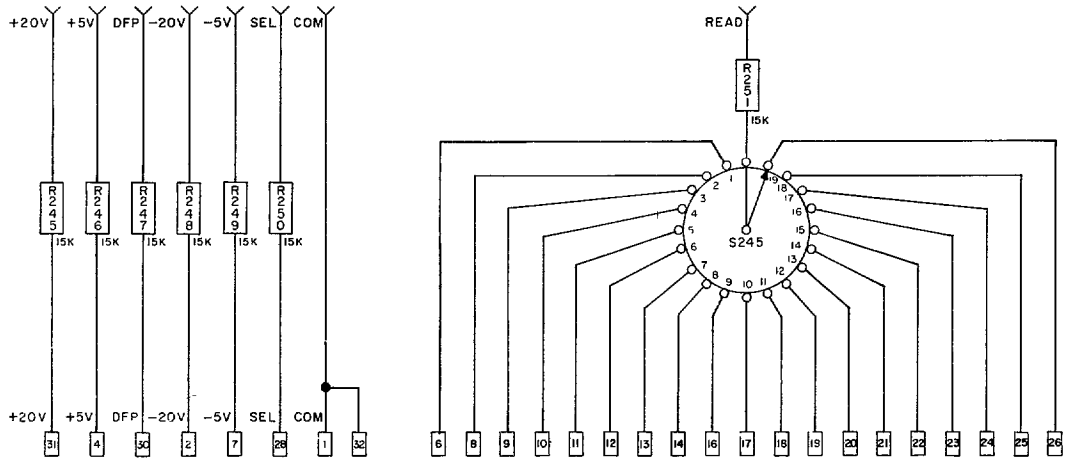
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

SCHEMATIC DIAGRAM

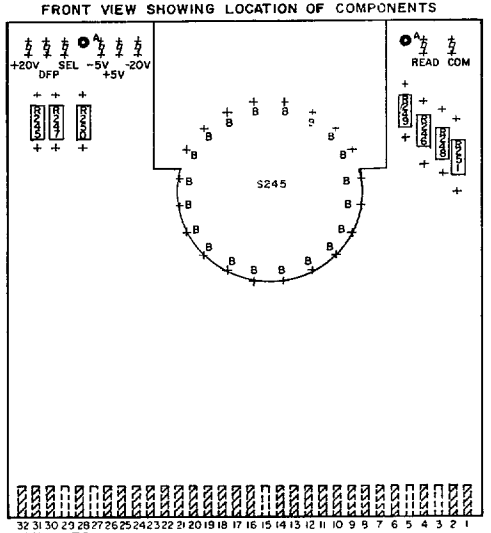
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
TEST CARD

FIRST MADE FOR STANDARD LINE
193X188AAG01



FRONT VIEW SHOWING LOCATION OF COMPONENTS



SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES 040 DIA.
EXCEPT THE HOLES
TABULATED BELOW

LOC	DIA	QUAN
A	157	2
B	062	20

NOTE 1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS

2 CROSS HATCHED TABS INDICATE TABS USED

3 CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	9428549AA004	2	0

REVISIONS

PRINTS TO

5QC(BK)5 5B(S)
5D(BK
5B(2
5(TS)5(
5(QC)

MADE BY G.R. Baker NOV. 24, 1970
ISSUED
APPROVALS E.R.S. K.W.P.
speed varidor
ERIE, PA.
DIV OR DEPT
36C762858AA
LOCATION CONT ON SHEET SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

36C762858AA
REV 0

GENERAL ELECTRIC

36C764129AA

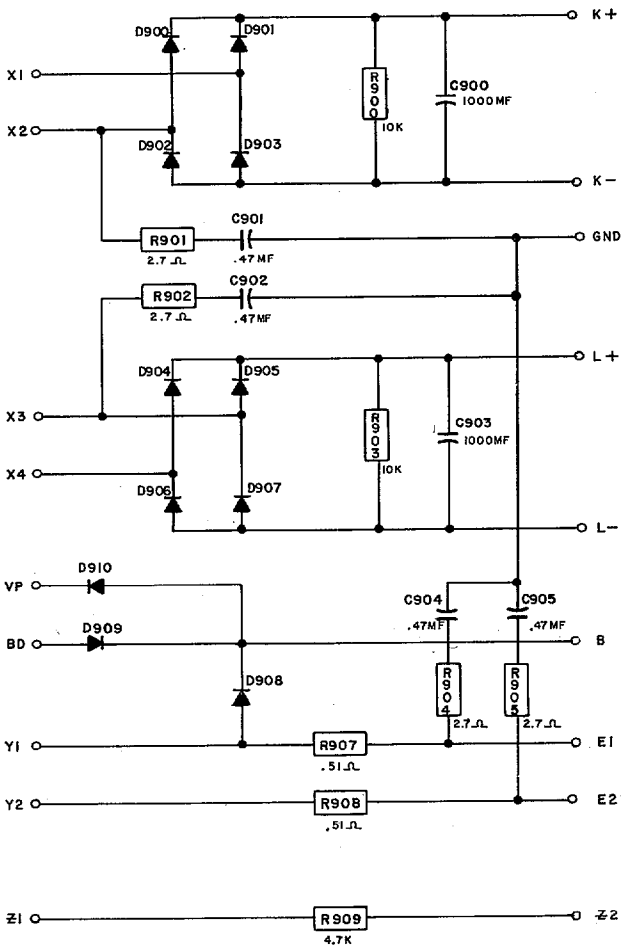
CONT ON SHEET | SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

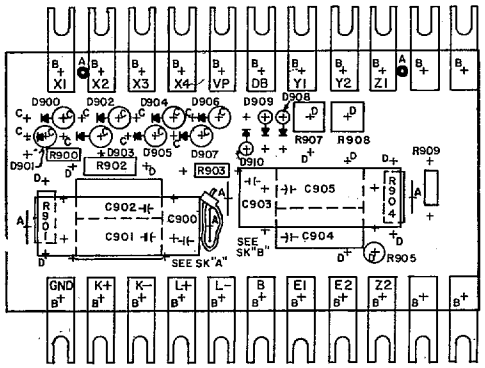
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
POWER INTERFACE
FIRST MADE FOR STANDARD LINE
193X370AA G01

SCHEMATIC DIAGRAM



FRONT VIEW SHOWING LOCATION OF COMPONENTS

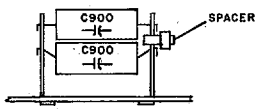


HOLE TABULATION

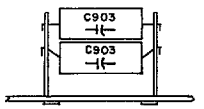
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	6
B	.101	22
C	.062	16
D	.052	12

CARD SIZE, $2.85^{\pm.020}$ X $5.130^{+.002}_{-.008}$



SKETCH "A"



SKETCH "B"

REV NO.
0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605049AA004	—	—

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5QC(BK)6	5 (S)
IT(BW)	5D(BK)
	5(TS)50
	5B(2)22
	JA(CD)
	5(QC)B

MADE BY R. Baba JUNE 20, 1973
ISSUED G. K. Baker Jr. FEB. 1, 1974

APPROVALS

SPEED VARIATOR DIV OR DEPT ERIE, PA. LOCATION 36C764129AA CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

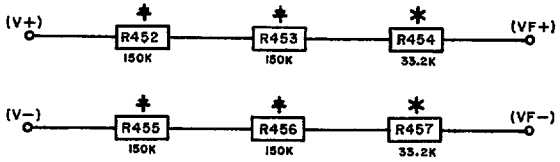
36C764140AA

SCHEMATIC DIAGRAM

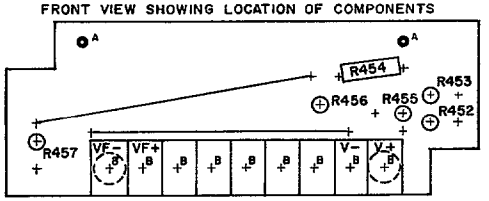
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
RESISTANCE ISOLATOR CARD

FIRST MADE FOR STANDARD LINE
193X263AAG03



FRONT VIEW SHOWING LOCATION OF COMPONENTS



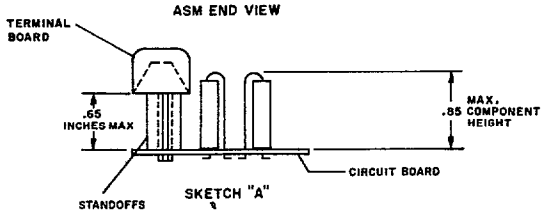
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	.157	2
B	.078	9

NOTE-1. CARD SIZE, 1.90 $^{+.000}_{-.015}$ X 5.130 $^{+.002}_{-.008}$

ASM END VIEW



SKETCH "A"

NOTES

1. ✱ 1% RESISTOR — 50 PPM
2. ✱ .2% RESISTOR — 50 PPM

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMEN	
		FRONT	BACK
FULL	36B605027AA004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varidator Department

PRINT

5QC(BK)5	JA(C
5B(2)22	5D(B
5(TS)50	IT(B
5(S)	5L(B
	5AE(B
	5(Q

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY J.R. Baker DEC. 17, 1973
ISSUED Paul Novak MARCH 26, 1974

APPROVALS

speed varidator
ERIE, PA.

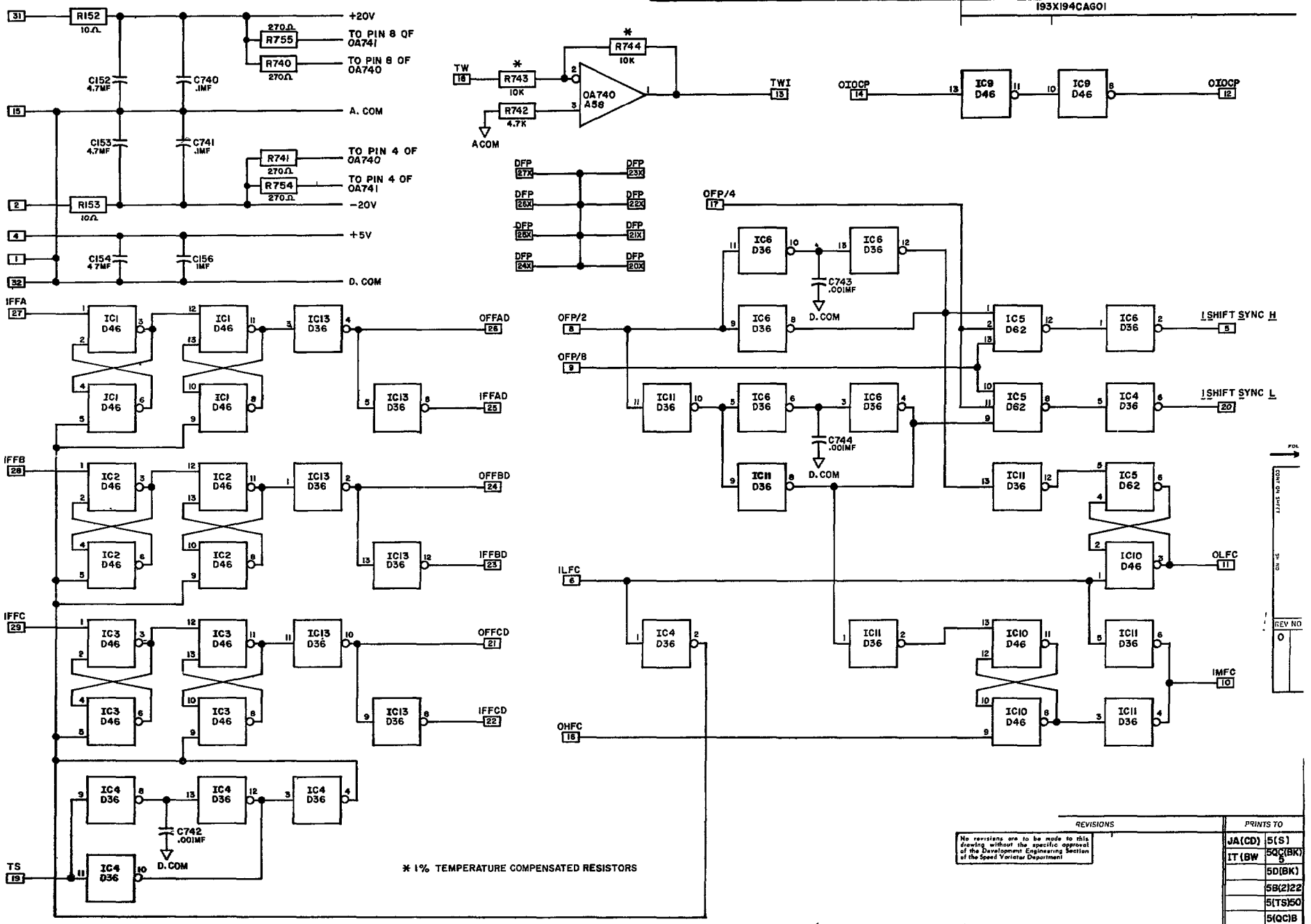
36C764140AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
PHASE SHIFT

FIRST MADE FOR STANDARD LINE
193X194CAG01



* 1% TEMPERATURE COMPENSATED RESISTORS

REV NO 0

REVISIONS

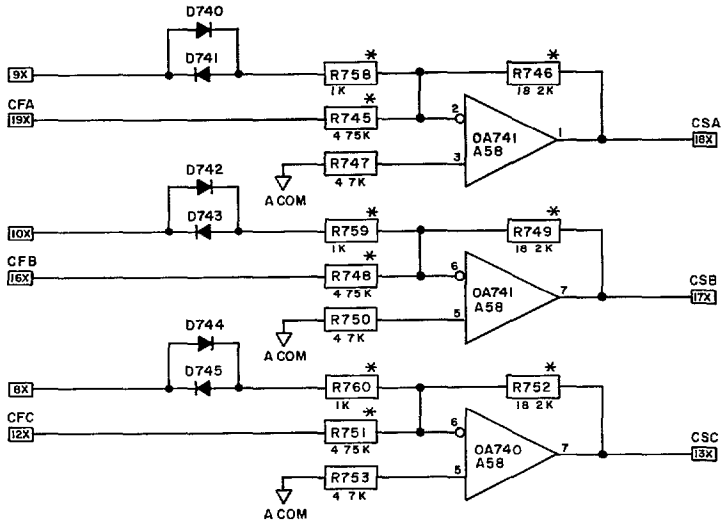
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

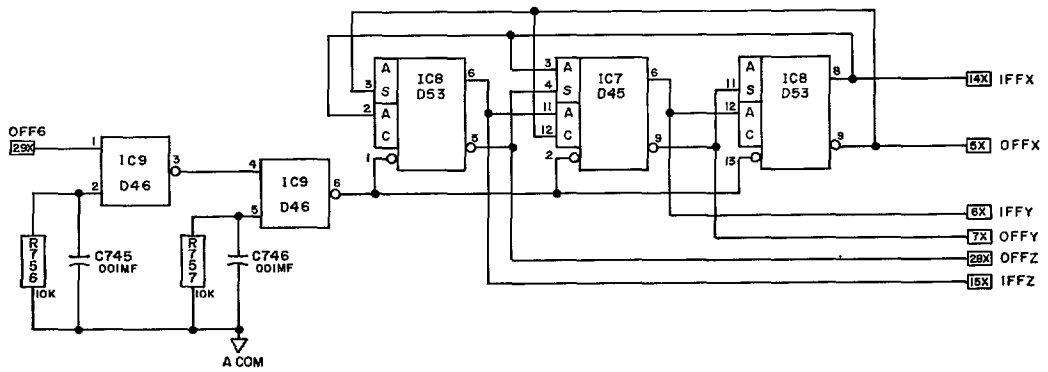
PRINTS TO

JA(CD)	5(S)
IT(BW	5QC(BK) 5
	5D(BK)
	5B(2)22
	5(TS)50
	5(QC)8

MADE BY O. Mastic AUG. 31, 1972
ISSUED D. Jones MARCH 15, 1974
APPROVALS
speed variator DIV OR DEPT
ERIE, PA. LOCATION
36C762893CA
CONT ON SHEET 2 SH NO 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY





JA(CD)	5B(2)22
IT(BW)	5D(BK)
	5(QC)18
	5QC(BK)5
	5(TS)50
	5(S)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY C Mattes AUG 31, 1972	APPROVALS	SPEED VARIATOR	36C762893CA
ISSUED D Monroe MARCH 15, 1974		ERIE, PA.	CONT'D SHEET 3 SH 2

GENERAL ELECTRIC

36C762893CA

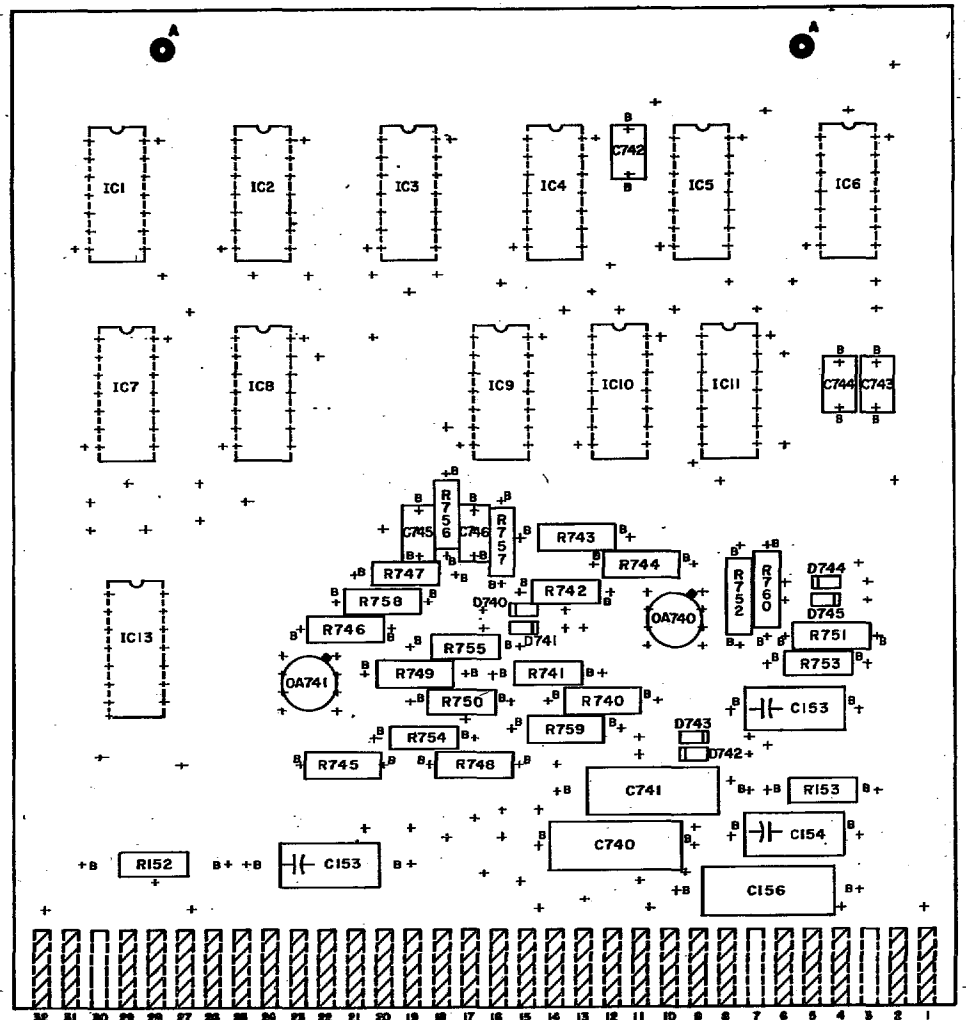
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: 71GA818 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
PHASE SHIFT CARD
FIRST MADE FOR STANDARD LINE
193X194CA601

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

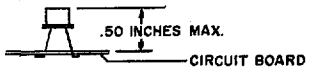
LOC.	DIA.	QUAN.
A	.157	2
B	.040	68



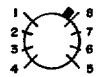
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.150^{+.002}_{-.006}$
4. THIS CARD HAS GOLD PLATED TABS ON BOTH SIDES. TABS 1 THRU 32 ARE LOCATED ON THE REVERSE SIDE. TABS 1X THRU 32X ARE LOCATED ON THE COMPONENT SIDE OF THE CARD. TAB 1X IS OPPOSITE TAB 1 AND ETC. THE TAB NUMBERS SHOWN ARE THOSE USED ON THIS CARD.
5. ALL OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

OP AMP (SIDE VIEW)

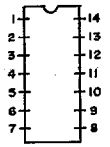


OP AMP LEAD SKETCH (TOP VIEW)



SKETCH FOR OA740, OA741

IC LEAD SKETCH (TOP VIEW)



SKETCH FOR IC1, IC2, IC3, IC4, IC5, IC6, IC7, IC8, IC9, IC10, IC11, IC13

REV NO.
0

2X	36A352957AC005	36A352958AC005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

IT(BW)	5B(2)22
JA(CD)	6D(BK)
	5(QC)B
	5QC(BK)5
	5(TS)50
	5(S)

DRAWN BY J. Mattei JUNE 11, 1973
ISSUED MARCH 15, 1974
APPROVED
SPEED VARIATOR PROD DIV. OR DEPT
ERIE, PA. USA LOCATION
36C762893CA COPY OR SHEET FL. SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY